Neural Computation -- 2019

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1. Toward Training Recurrent Neural Networks for Lifelong Learning

Shagun Sodhani, Sarath Chandar, Yoshua Bengio

https://doi.org/10.1162/neco_a_01246 | pp. 1-35

Catastrophic forgetting and capacity saturation are the central challenges of any parametric lifelong learning system. In this work, we study these challenges in the context of sequential supervised learning with an emphasis on recurrent neural networks. To evaluate the models in the lifelong learning setting, we propose a curriculum-based, simple, and intuitive benchmark where the models are trained on tasks with increasing levels of difficulty. To measure the impact of catastrophic forgetting, the model is tested on all the previous tasks as it completes any task. As a step toward developing true lifelong learning systems, we unify gradient episodic memory (a catastrophic forgetting alleviation approach) and Net2Net (a capacity expansion approach). Both models are proposed in the context of feedforward networks, and we evaluate the feasibility of using them for recurrent networks. Evaluation on the proposed benchmark shows that the unified model is more suitable than the constituent models for lifelong learning setting.

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2. A Continuous-Time Analysis of Distributed Stochastic Gradient

Nicholas M. Boffi, Jean-Jacques E. Slotine

https://doi.org/10.1162/neco_a_01248 | pp. 36-96

We analyze the effect of synchronization on distributed stochastic gradient algorithms. By exploiting an analogy with dynamical models of biological quorum sensing, where synchronization between agents is induced through communication with a common signal, we quantify how synchronization can significantly reduce the magnitude of the noise felt by the individual distributed agents and their spatial mean. This noise reduction is in turn associated with a reduction in the smoothing of the loss function imposed by the stochastic gradient approximation. Through simulations on model nonconvex objectives, we demonstrate that coupling can stabilize higher noise levels and improve convergence. We provide a convergence analysis for strongly convex functions by deriving a bound on the expected deviation of the spatial mean of the agents from the global minimizer for an algorithm based on quorum sensing, the same algorithm with momentum, and the elastic averaging SGD (EASGD) algorithm. We discuss extensions to new algorithms that allow each agent to broadcast its current measure of success and shape the collective computation accordingly. We supplement our theoretical analysis with numerical experiments on convolutional neural networks trained on the CIFAR-10 data set, where we note a surprising regularizing property of EASGD even when applied to the non-distributed case. This observation suggests alternative second-order in time algorithms for nondistributed optimization that are competitive with momentum methods.

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3. On Kernel Method–Based Connectionist Models and Supervised Deep Learning Without Backpropagation

Shiyu Duan, Shujian Yu, Yunmei Chen, José C. Pr■■ncipe

https://doi.org/10.1162/neco_a_01250 | pp. 97-135

We propose a novel family of connectionist models based on kernel machines and consider the problem of learning layer by layer a compositional hypothesis class (i.e., a feedforward, multilayer architecture) in a supervised setting. In terms of the models, we present a principled method to “kernelize” (partly or completely) any neural network (NN). With this method, we obtain a counterpart of any given NN that is powered by kernel machines instead of neurons. In terms of learning, when learning a feedforward deep architecture in a supervised setting, one needs to train all the components simultaneously using backpropagation (BP) since there are no explicit targets for the hidden layers (Rumelhart, Hinton, & Williams, 1986). We consider without loss of generality the two-layer case and present a general framework that explicitly characterizes a target for the hidden layer that is optimal for minimizing the objective function of the network. This characterization then makes possible a purely greedy training scheme that learns one layer at a time, starting from the input layer. We provide instantiations of the abstract framework under certain architectures and objective functions. Based on these instantiations, we present a layer-wise training algorithm for an [Formula: see text]-layer feedforward network for classification, where [Formula: see text] can be arbitrary. This algorithm can be given an intuitive geometric interpretation that makes the learning dynamics transparent. Empirical results are provided to complement our theory. We show that the kernelized networks, trained layer-wise, compare favorably with classical kernel machines as well as other connectionist models trained by BP. We also visualize the inner workings of the greedy kernelized models to validate our claim on the transparency of the layer-wise algorithm.

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4. Storing Object-Dependent Sparse Codes in a Willshaw Associative Network

Luis Sa-Couto, Andreas Wichert

https://doi.org/10.1162/neco_a_01243 | pp. 136-152

Willshaw networks are single-layered neural networks that store associations between binary vectors. Using only binary weights, these networks can be implemented efficiently to store large numbers of patterns and allow for fault-tolerant recovery of those patterns from noisy cues. However, this is only the case when the involved codes are sparse and randomly generated. In this letter, we use a recently proposed approach that maps visual patterns into informative binary features. By doing so, we manage to transform MNIST handwritten digits into well-distributed codes that we then store in a Willshaw network in autoassociation. We perform experiments with both noisy and noiseless cues and verify a tenuous impact on the recovered pattern's relevant information. More specifically, we were able to perform retrieval after filling the memory to several factors of its number of units while preserving the information of the class to which the pattern belongs.

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5. A Robust Model of Gated Working Memory

Anthony Strock, Xavier Hinaut, Nicolas P. Rougier

https://doi.org/10.1162/neco_a_01249 | pp. 153-181

Gated working memory is defined as the capacity of holding arbitrary information at any time in order to be used at a later time. Based on electrophysiological recordings, several computational models have tackled the problem using dedicated and explicit mechanisms. We propose instead to consider an implicit mechanism based on a random recurrent neural network. We introduce a robust yet simple reservoir model

of gated working memory with instantaneous updates. The model is able to store an arbitrary real value at random time over an extended period of time. The dynamics of the model is a line attractor that learns to exploit reentry and a nonlinearity during the training phase using only a few representative values. A deeper study of the model shows that there is actually a large range of hyperparameters for which the results hold (e.g., number of neurons, sparsity, global weight scaling) such that any large enough population, mixing excitatory and inhibitory neurons, can quickly learn to realize such gated working memory. In a nutshell, with a minimal set of hypotheses, we show that we can have a robust model of working memory. This suggests this property could be an implicit property of any random population, that can be acquired through learning. Furthermore, considering working memory to be a physically open but functionally closed system, we give account on some counterintuitive electrophysiological recordings.

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6. An FPGA Implementation of Deep Spiking Neural Networks for Low-Power and Fast Classification

Xiping Ju, Biao Fang, Rui Yan, Xiaoliang Xu, Huajin Tang

https://doi.org/10.1162/neco_a_01245 | pp. 182-204

A spiking neural network (SNN) is a type of biological plausibility model that performs information processing based on spikes. Training a deep SNN effectively is challenging due to the nondifferention of spike signals. Recent advances have shown that high-performance SNNs can be obtained by converting convolutional neural networks (CNNs). However, the large-scale SNNs are poorly served by conventional architectures due to the dynamic nature of spiking neurons. In this letter, we propose a hardware architecture to enable efficient implementation of SNNs. All layers in the network are mapped on one chip so that the computation of different time steps can be done in parallel to reduce latency. We propose new spiking max-pooling method to reduce computation complexity. In addition, we apply approaches based on shift register and coarsely grained parallels to accelerate convolution operation. We also investigate the effect of different encoding methods on SNN accuracy. Finally, we validate the hardware architecture on the Xilinx Zynq ZCU102. The experimental results on the MNIST data set show that it can achieve an accuracy of 98.94% with eight-bit quantized weights. Furthermore, it achieves 164 frames per second (FPS) under 150 MHz clock frequency and obtains 41[Formula: see text] speed-up compared to CPU implementation and 22 times lower power than GPU implementation.

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7. Iterative Retrieval and Block Coding in Autoassociative and Heteroassociative Memory

Andreas Knoblauch, Günther Palm

https://doi.org/10.1162/neco_a_01247 | pp. 205-260

Neural associative memories (NAM) are perceptron-like single-layer networks with fast synaptic learning typically storing discrete associations between pairs of neural activity patterns. Gripon and Berrou (2011) investigated NAM employing block coding, a particular sparse coding method, and reported a significant increase in storage capacity. Here we verify and extend their results for both heteroassociative and recurrent autoassociative networks. For this we provide a new analysis of iterative retrieval in finite autoassociative and heteroassociative networks that allows estimating storage capacity for random and block patterns. Furthermore, we have implemented various retrieval algorithms for block coding and compared them in

simulations to our theoretical results and previous simulation data. In good agreement of theory and experiments, we find that finite networks employing block coding can store significantly more memory patterns. However, due to the reduced information per block pattern, it is not possible to significantly increase stored information per synapse. Asymptotically, the information retrieval capacity converges to the known limits [Formula: see text] and [Formula: see text] also for block coding. We have also implemented very large recurrent networks up to [Formula: see text] neurons, showing that maximal capacity [Formula: see text] bit per synapse occurs for finite networks having a size [Formula: see text] similar to cortical macrocolumns.

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8. Optimal Sampling of Parametric Families: Implications for Machine Learning

Adrian Huber, Jithendar Anumula, Shih■Chii Liu

https://doi.org/10.1162/neco_a_01251 | pp. 261-279

It is well known in machine learning that models trained on a training set generated by a probability distribution function perform far worse on test sets generated by a different probability distribution function. In the limit, it is feasible that a continuum of probability distribution functions might have generated the observed test set data; a desirable property of a learned model in that case is its ability to describe most of the probability distribution functions from the continuum equally well. This requirement naturally leads to sampling methods from the continuum of probability distribution functions that lead to the construction of optimal training sets. We study the sequential prediction of Ornstein-Uhlenbeck processes that form a parametric family. We find empirically that a simple deep network trained on optimally constructed training sets using the methods described in this letter can be robust to changes in the test set distribution.

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9. Face Representations via Tensorfaces of Various Complexities

Sidney R. Lehky, Anh Huy Phan, Andrzej Cichocki, Keiji Tanaka

https://doi.org/10.1162/neco_a_01258 | pp. 281-329

Neurons selective for faces exist in humans and monkeys. However, characteristics of face cell receptive fields are poorly understood. In this theoretical study, we explore the effects of complexity, defined as algorithmic information (Kolmogorov complexity) and logical depth, on possible ways that face cells may be organized. We use tensor decompositions to decompose faces into a set of components, called tensorfaces, and their associated weights, which can be interpreted as model face cells and their firing rates. These tensorfaces form a high-dimensional representation space in which each tensorface forms an axis of the space. A distinctive feature of the decomposition algorithm is the ability to specify tensorface complexity. We found that low-complexity tensorfaces have blob-like appearances crudely approximating faces, while high-complexity tensorfaces appear clearly face-like. Low-complexity tensorfaces require a larger population to reach a criterion face reconstruction error than medium- or high-complexity tensorfaces, and thus are inefficient by that criterion. Low-complexity tensorfaces, however, generalize better when representing statistically novel faces, which are faces falling beyond the distribution of face description parameters found in the tensorface training set. The degree to which face representations are parts based or global forms a continuum as a function of tensorface complexity, with low and medium tensorfaces being more parts based. Given the computational load imposed in creating high-complexity face cells (in the form of

algorithmic information and logical depth) and in the absence of a compelling advantage to using high-complexity cells, we suggest face representations consist of a mixture of low- and medium-complexity face cells.

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10. Transition Scale-Spaces: A Computational Theory for the Discretized Entorhinal Cortex

Nicolai Waniek

https://doi.org/10.1162/neco_a_01255 | pp. 330-394

Although hippocampal grid cells are thought to be crucial for spatial navigation, their computational purpose remains disputed. Recently, they were proposed to represent spatial transitions and convey this knowledge downstream to place cells. However, a single scale of transitions is insufficient to plan long goal-directed sequences in behaviorally acceptable time. Here, a scale-space data structure is suggested to optimally accelerate retrievals from transition systems, called transition scale-space (TSS). Remaining exclusively on an algorithmic level, the scale increment is proved to be ideally [Formula: see text] for biologically plausible receptive fields. It is then argued that temporal buffering is necessary to learn the scale-space online. Next, two modes for retrieval of sequences from the TSS are presented: top down and bottom up. The two modes are evaluated in symbolic simulations (i.e., without biologically plausible spiking neurons). Additionally, a TSS is used for short-cut discovery in a simulated Morris water maze. Finally, the results are discussed in depth with respect to biological plausibility, and several testable predictions are derived. Moreover, relations to other grid cell models, multiresolution path planning, and scale-space theory are highlighted. Summarized, reward-free transition encoding is shown here, in a theoretical model, to be compatible with the observed discretization along the dorso-ventral axis of the medial entorhinal cortex. Because the theoretical model generalizes beyond navigation, the TSS is suggested to be a general-purpose cortical data structure for fast retrieval of sequences and relational knowledge. Source code for all simulations presented in this paper can be found at https://github.com/rochus/transitionscalespace .

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11. From Synaptic Interactions to Collective Dynamics in Random Neuronal Networks Models: Critical Role of Eigenvectors and Transient Behavior

E. Gudowska-Nowak, M. A. Nowak, D. R. Chialvo, J. K. Ochab, W. Tarnowski

https://doi.org/10.1162/neco_a_01253 | pp. 395-423

The study of neuronal interactions is at the center of several big collaborative neuroscience projects (including the Human Connectome Project, the Blue Brain Project, and the Brainome) that attempt to obtain a detailed map of the entire brain. Under certain constraints, mathematical theory can advance predictions of the expected neural dynamics based solely on the statistical properties of the synaptic interaction matrix. This work explores the application of free random variables to the study of large synaptic interaction matrices. Besides recovering in a straightforward way known results on eigenspectra in types of models of neural networks proposed by Rajan and Abbott (2006), we extend them to heavy-tailed distributions of interactions. More important, we analytically derive the behavior of eigenvector overlaps, which determine the stability of the spectra. We observe that on imposing the neuronal excitation/inhibition balance, despite the eigenvalues remaining unchanged, their stability dramatically decreases due to the strong nonorthogonality of associated eigenvectors. This leads us to the conclusion that understanding the

temporal evolution of asymmetric neural networks requires considering the entangled dynamics of both eigenvectors and eigenvalues, which might bear consequences for learning and memory processes in these models. Considering the success of free random variables theory in a wide variety of disciplines, we hope that the results presented here foster the additional application of these ideas in the area of brain sciences.

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12. Synaptic Scaling Improves the Stability of Neural Mass Models Capable of Simulating Brain Plasticity

Jure Demšar, Rob Forsyth

https://doi.org/10.1162/neco_a_01257 | pp. 424-446

Neural mass models offer a way of studying the development and behavior of large-scale brain networks through computer simulations. Such simulations are currently mainly research tools, but as they improve, they could soon play a role in understanding, predicting, and optimizing patient treatments, particularly in relation to effects and outcomes of brain injury. To bring us closer to this goal, we took an existing state-of-the-art neural mass model capable of simulating connection growth through simulated plasticity processes. We identified and addressed some of the model's limitations by implementing biologically plausible mechanisms. The main limitation of the original model was its instability, which we addressed by incorporating a representation of the mechanism of synaptic scaling and examining the effects of optimizing parameters in the model. We show that the updated model retains all the merits of the original model, while being more stable and capable of generating networks that are in several aspects similar to those found in real brains.

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13. Scaled Coupled Norms and Coupled Higher-Order Tensor Completion

Kishan Wimalawarne, Makoto Yamada, Hiroshi Mamitsuka

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has been proposed as a convex solution to coupled tensor completion. Coupled norms have been designed by combining low-rank inducing tensor norms with the matrix trace norm. Though coupled norms have shown good performances, they have two major limitations: they do not have a method to control the regularization of coupled modes and uncoupled modes, and they are not optimal for couplings among higher-order tensors. In this letter, we propose a method that scales the regularization of coupled components against uncoupled components to properly induce the low-rankness on the coupled mode. We also propose coupled norms for higher-order tensors by combining the square norm to coupled norms. Using the excess risk-bound analysis, we demonstrate that our proposed methods lead to lower risk bounds compared to existing coupled norms. We demonstrate the robustness of our methods through simulation and real-data experiments.

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14. Gradient Descent with Identity Initialization Efficiently Learns Positive-Definite Linear Transformations by Deep Residual Networks

Peter L. Bartlett, David P. Helmbold, Philip M. Long

https://doi.org/10.1162/neco_a_01164 | pp. 477-502

We analyze algorithms for approximating a function [Formula: see text] mapping [Formula: see text] to [Formula: see text] using deep linear neural networks, that is, that learn a function [Formula: see text] parameterized by matrices [Formula: see text] and defined by [Formula: see text]. We focus on algorithms that learn through gradient descent on the population quadratic loss in the case that the distribution over the inputs is isotropic. We provide polynomial bounds on the number of iterations for gradient descent to approximate the least-squares matrix [Formula: see text], in the case where the initial hypothesis [Formula: see text] has excess loss bounded by a small enough constant. We also show that gradient descent fails to converge for [Formula: see text] whose distance from the identity is a larger constant, and we show that some forms of regularization toward the identity in each layer do not help. If [Formula: see text] is symmetric positive definite, we show that an algorithm that initializes [Formula: see text] learns an [Formula: see text]-approximation of [Formula: see text] using a number of updates polynomial in [Formula: see text], the condition number of [Formula: see text], and [Formula: see text]. In contrast, we show that if the least-squares matrix [Formula: see text] is symmetric and has a negative eigenvalue, then all members of a class of algorithms that perform gradient descent with identity initialization, and optionally regularize toward the identity in each layer, fail to converge. We analyze an algorithm for the case that [Formula: see text] satisfies [Formula: see text] for all [Formula: see text] but may not be symmetric. This algorithm uses two regularizers: one that maintains the invariant [Formula: see text] for all [Formula: see text] and the other that “balances” [Formula: see text] so that they have the same singular values.

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15. Improving Generalization via Attribute Selection on Out-of-the-Box Data

Xiaofeng Xu, Ivor W. Tsang, Chuancai Liu

https://doi.org/10.1162/neco_a_01256 | pp. 485-514

Zero-shot learning (ZSL) aims to recognize unseen objects (test classes) given some other seen objects (training classes) by sharing information of attributes between different objects. Attributes are artificially annotated for objects and treated equally in recent ZSL tasks. However, some inferior attributes with poor predictability or poor discriminability may have negative impacts on the ZSL system performance. This letter first derives a generalization error bound for ZSL tasks. Our theoretical analysis verifies that selecting the subset of key attributes can improve the generalization performance of the original ZSL model, which uses all the attributes. Unfortunately, previous attribute selection methods have been conducted based on the seen data, and their selected attributes have poor generalization capability to the unseen data, which is unavailable in the training stage of ZSL tasks. Inspired by learning from pseudo-relevance feedback, this letter introduces out-of-the-box data-pseudo-data generated by an attribute-guided generative model-to mimic the unseen data. We then present an iterative attribute selection (IAS) strategy that iteratively selects key attributes based on the out-of-the-box data. Since the distribution of the generated out-of-the-box data is similar to that of the test data, the key attributes selected by IAS can be effectively generalized to test data. Extensive experiments demonstrate that IAS can significantly improve existing attribute-based ZSL methods and achieve state-of-the-art performance.

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16. Forgetting Memories and Their Attractiveness

Enzo Marinari

https://doi.org/10.1162/neco_a_01162 | pp. 503-516

We study numerically the memory that forgets, introduced in 1986 by Parisi by bounding the synaptic strength, with a mechanism that avoids confusion; allows remembering the pattern learned more recently; and has a physiologically very well-defined meaning. We analyze a number of features of this learning for a finite number of neurons and finite number of patterns. We discuss how the system behaves in the large but finite [Formula: see text] limit. We analyze the basin of attraction of the patterns that have been learned, and we show that it is exponentially small in the age of the pattern.

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17. Scalable and Flexible Unsupervised Feature Selection

Haojie Hu, Rong Wang, Xiaojun Yang, Feiping Nie

https://doi.org/10.1162/neco_a_01163 | pp. 517-537

Recently, graph-based unsupervised feature selection algorithms (GUFS) have been shown to efficiently handle prevalent high-dimensional unlabeled data. One common drawback associated with existing graph-based approaches is that they tend to be time-consuming and in need of large storage, especially when faced with the increasing size of data. Research has started using anchors to accelerate graph-based learning model for feature selection, while the hard linear constraint between the data matrix and the lower-dimensional representation is usually overstrict in many applications. In this letter, we propose a flexible linearization model with anchor graph and [Formula: see text]-norm regularization, which can deal with large-scale data sets and improve the performance of the existing anchor-based method. In addition, the anchor-based graph Laplacian is constructed to characterize the manifold embedding structure by means of a parameter-free adaptive neighbor assignment strategy. An efficient iterative algorithm is developed to address the optimization problem, and we also prove the convergence of the algorithm. Experiments on several public data sets demonstrate the effectiveness and efficiency of the method we propose.

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18. State-Space Representations of Deep Neural Networks

Michael Hauser, Sean Gunn, Samer S. Saab, Asok Ray

https://doi.org/10.1162/neco_a_01165 | pp. 538-554

This letter deals with neural networks as dynamical systems governed by finite difference equations. It shows that the introduction of [Formula: see text]-many skip connections into network architectures, such as residual networks and additive dense networks, defines [Formula: see text]th order dynamical equations on the layer-wise transformations. Closed-form solutions for the state-space representations of general [Formula: see text]th order additive dense networks, where the concatenation operation is replaced by addition, as well as [Formula: see text]th order smooth networks, are found. The developed provision endows deep neural networks with an algebraic structure. Furthermore, it is shown that imposing [Formula: see text]th order smoothness on network architectures with [Formula: see text]-many nodes per layer increases the state-space dimension by a multiple of [Formula: see text], and so the effective embedding dimension of the data manifold by the neural network is [Formula: see text]-many dimensions. It follows that network architectures of these types reduce the number of parameters needed to maintain the same embedding dimension by a factor of [Formula: see text] when compared to an equivalent first-order, residual network. Numerical simulations and experiments on CIFAR10, SVHN, and MNIST have been conducted to help understand the developed theory and efficacy of the proposed concepts.

19. Advancing System Performance with Redundancy: From Biological to Artificial Designs

Anh Tuan Nguyen, Jian Xu, Diu Khue Luu, Qi Zhao, Zhi Yang

https://doi.org/10.1162/neco_a_01166 | pp. 555-573

Redundancy is a fundamental characteristic of many biological processes such as those in the genetic, visual, muscular, and nervous systems, yet its driven mechanism has not been fully comprehended. Until recently, the only understanding of redundancy is as a mean to attain fault tolerance, which is reflected in the design of many man-made systems. On the contrary, our previous work on redundant sensing (RS) has demonstrated an example where redundancy can be engineered solely for enhancing accuracy and precision. The design was inspired by the binocular structure of human vision, which we believe may share a similar operation. In this letter, we present a unified theory describing how such utilization of redundancy is feasible through two complementary mechanisms: representational redundancy (RPR) and entangled redundancy (ETR). We also point out two additional examples where our new understanding of redundancy can be applied to justify a system's superior performance. One is the human musculoskeletal system (HMS), a biological instance, and the other is the deep residual neural network (ResNet), an artificial counterpart. We envision that our theory would provide a framework for the future development of bio-inspired redundant artificial systems, as well as assist studies of the fundamental mechanisms governing various biological processes.

20. A Novel Optimization Framework to Improve the Computational Cost of Muscle Activation Prediction for a Neuromusculoskeletal System

Seyed Mohammad Ali Rahmati, Mostafa Rostami, Alireza Karimi

https://doi.org/10.1162/neco_a_01167 | pp. 574-595

The high computational cost (CC) of neuromusculoskeletal modeling is usually considered a serious barrier in clinical applications. Different approaches have been developed to lessen CC and amplify the accuracy of muscle activation prediction based on forward and inverse analyses by applying different optimization algorithms. This study is aimed at proposing two novel approaches, inverse muscular dynamics with inequality constraints (IMDIC) and inverse-forward muscular dynamics with inequality constraints (IFMDIC), not only to reduce CC but also to amend the computational errors compared to the well-known approach of extended inverse dynamics (EID). To do that, the equality constraints of optimization problem, which are computationally tough to satisfy, are replaced by inequality constraints, which are easier to satisfy. To verify the practical application of the proposed approaches, the muscle activations of the lower limbs during the half of a gait cycle are quantified. The simulation results of the optimal muscle activations are then compared to the experimental ones. The results reveal that IMDIC requires less CC (87.5%) compared to EID. In addition, CC of IMDIC was about 33.3% improved by the application of IFMDIC. The findings of this study suggest that although the novel approach of IFMDIC decreases CC compared to IMDIC, the convergence of its results is very sensitive to the primary guess of the optimization variables.

21. A Distributed Framework for the Construction of Transport Maps

Diego Mesa, Justin Tantiongloc, Marcela Mendoza, Sanggyun Kim, Todd P. Coleman
https://doi.org/10.1162/neco_a_01172 | pp. 613-652

The need to reason about uncertainty in large, complex, and multimodal data sets has become increasingly common across modern scientific environments. The ability to transform samples from one distribution [Formula: see text] to another distribution [Formula: see text] enables the solution to many problems in machine learning (e.g., Bayesian inference, generative modeling) and has been actively pursued from theoretical, computational, and application perspectives across the fields of information theory, computer science, and biology. Performing such transformations in general still leads to computational difficulties, especially in high dimensions. Here, we consider the problem of computing such “measure transport maps” with efficient and parallelizable methods. Under the mild assumptions that [Formula: see text] need not be known but can be sampled from and that the density of [Formula: see text] is known up to a proportionality constant, and that [Formula: see text] is log-concave, we provide in this work a convex optimization problem pertaining to relative entropy minimization. We show how an empirical minimization formulation and polynomial chaos map parameterization can allow for learning a transport map between [Formula: see text] and [Formula: see text] with distributed and scalable methods. We also leverage findings from nonequilibrium thermodynamics to represent the transport map as a composition of simpler maps, each of which is learned sequentially with a transport cost regularized version of the aforementioned problem formulation. We provide examples of our framework within the context of Bayesian inference for the Boston housing data set and generative modeling for handwritten digit images from the MNIST data set.

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22. Biologically Realistic Mean-Field Models of Conductance-Based Networks of Spiking Neurons with Adaptation

Matteo di Volo, Alberto Romagnoni, Cristiano Capone, Alain Destexhe
https://doi.org/10.1162/neco_a_01173 | pp. 653-680

Accurate population models are needed to build very large-scale neural models, but their derivation is difficult for realistic networks of neurons, in particular when nonlinear properties are involved, such as conductance-based interactions and spike-frequency adaptation. Here, we consider such models based on networks of adaptive exponential integrate-and-fire excitatory and inhibitory neurons. Using a master equation formalism, we derive a mean-field model of such networks and compare it to the full network dynamics. The mean-field model is capable of correctly predicting the average spontaneous activity levels in asynchronous irregular regimes similar to in vivo activity. It also captures the transient temporal response of the network to complex external inputs. Finally, the mean-field model is also able to quantitatively describe regimes where high- and low-activity states alternate (up-down state dynamics), leading to slow oscillations. We conclude that such mean-field models are biologically realistic in the sense that they can capture both spontaneous and evoked activity, and they naturally appear as candidates to build very large-scale models involving multiple brain areas.

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23. Estimating Scale-Invariant Future in Continuous Time

Zoran Tiganj, Samuel J. Gershman, Per B. Sederberg, Marc W. Howard
https://doi.org/10.1162/neco_a_01171 | pp. 681-709

Natural learners must compute an estimate of future outcomes that follow from a stimulus in continuous time. Widely used reinforcement learning algorithms discretize continuous time and estimate either transition functions from one step to the next (model-based algorithms) or a scalar value of exponentially discounted future reward using the Bellman equation (model-free algorithms). An important drawback of model-based algorithms is that computational cost grows linearly with the amount of time to be simulated. An important drawback of model-free algorithms is the need to select a timescale required for exponential discounting. We present a computational mechanism, developed based on work in psychology and neuroscience, for computing a scale-invariant timeline of future outcomes. This mechanism efficiently computes an estimate of inputs as a function of future time on a logarithmically compressed scale and can be used to generate a scale-invariant power-law-discounted estimate of expected future reward. The representation of future time retains information about what will happen when. The entire timeline can be constructed in a single parallel operation that generates concrete behavioral and neural predictions. This computational mechanism could be incorporated into future reinforcement learning algorithms.

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24. Deconstructing Odorant Identity via Primacy in Dual Networks

Daniel Kepple, Hamza Giaffar, Dmitry Rinberg, Alexei A. Koulakov

https://doi.org/10.1162/neco_a_01175 | pp. 710-737

In the olfactory system, odor percepts retain their identity despite substantial variations in concentration, timing, and background. We study a novel strategy for encoding intensity-invariant stimulus identity that is based on representing relative rather than absolute values of stimulus features. For example, in what is known as the primacy coding model, odorant identities are represented by the conditions that some odorant receptors are activated more strongly than others. Because, in this scheme, odorant identity depends only on the relative amplitudes of olfactory receptor responses, identity is invariant to changes in both intensity and monotonic nonlinear transformations of its neuronal responses. Here we show that sparse vectors representing odorant mixtures can be recovered in a compressed sensing framework via elastic net loss minimization. In the primacy model, this minimization is performed under the constraint that some receptors respond to a given odorant more strongly than others. Using duality transformation, we show that this constrained optimization problem can be solved by a neural network whose Lyapunov function represents the dual Lagrangian and whose neural responses represent the Lagrange coefficients of primacy and other constraints. The connectivity in such a dual network resembles known features of connectivity in olfactory circuits. We thus propose that networks in the piriform cortex implement dual computations to compute odorant identity with the sparse activities of individual neurons representing Lagrange coefficients. More generally, we propose that sparse neuronal firing rates may represent Lagrange multipliers, which we call the dual brain hypothesis. We show such a formulation is well suited to solve problems with multiple interacting relative constraints.

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25. Filtering Compensation for Delays and Prediction Errors during Sensorimotor Control

Frédéric Crevecoeur, M. Gevers

https://doi.org/10.1162/neco_a_01170 | pp. 738-764

Compensating for sensorimotor noise and for temporal delays has been identified as a major function of the nervous system. Although these aspects have often been described separately in the frameworks of optimal cue combination or motor prediction during movement planning, control-theoretic models suggest that these two operations are performed simultaneously, and mounting evidence supports that motor commands are based on sensory predictions rather than sensory states. In this letter, we study the benefit of state estimation for predictive sensorimotor control. More precisely, we combine explicit compensation for sensorimotor delays and optimal estimation derived in the context of Kalman filtering. We show, based on simulations of human-inspired eye and arm movements, that filtering sensory predictions improves the stability margin of the system against prediction errors due to low-dimensional predictions or to errors in the delay estimate. These simulations also highlight that prediction errors qualitatively account for a broad variety of movement disorders typically associated with cerebellar dysfunctions. We suggest that adaptive filtering in cerebellum, instead of often-assumed feedforward predictions, may achieve simple compensation for sensorimotor delays and support stable closed-loop control of movements.

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26. Gated Orthogonal Recurrent Units: On Learning to Forget

Li Jing, Ça■lar Gülçehre, John Peurifoy, Yichen Shen, Max Tegmark, Marin Solja■i■, Yoshua Bengio

https://doi.org/10.1162/neco_a_01174 | pp. 765-783

We present a novel recurrent neural network (RNN)-based model that combines the remembering ability of unitary evolution RNNs with the ability of gated RNNs to effectively forget redundant or irrelevant information in its memory. We achieve this by extending restricted orthogonal evolution RNNs with a gating mechanism similar to gated recurrent unit RNNs with a reset gate and an update gate. Our model is able to outperform long short-term memory, gated recurrent units, and vanilla unitary or orthogonal RNNs on several long-term-dependency benchmark tasks. We empirically show that both orthogonal and unitary RNNs lack the ability to forget. This ability plays an important role in RNNs. We provide competitive results along with an analysis of our model on many natural sequential tasks, including question answering, speech spectrum prediction, character-level language modeling, and synthetic tasks that involve long-term dependencies such as algorithmic, denoising, and copying tasks.

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27. Decreasing the Size of the Restricted Boltzmann Machine

Yohei Saito, Takuya Kato

https://doi.org/10.1162/neco_a_01176 | pp. 784-805

In this letter, we propose a method to decrease the number of hidden units of the restricted Boltzmann machine while avoiding a decrease in the performance quantified by the Kullback-Leibler divergence. Our algorithm is then demonstrated by numerical simulations.

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28. Multiclass Alpha Integration of Scores from Multiple Classifiers

Gonzalo Safont, Addisson Salazar, Luis Vergara

https://doi.org/10.1162/neco_a_01169 | pp. 806-825

Alpha integration methods have been used for integrating stochastic models and fusion in the context of detection (binary classification). Our work proposes separated score integration (SSI), a new method based on alpha integration to perform soft fusion of scores in multiclass classification problems, one of the most common problems in automatic classification. Theoretical derivation is presented to optimize the parameters of this method to achieve the least mean squared error (LMSE) or the minimum probability of error (MPE). The proposed alpha integration method was tested on several sets of simulated and real data. The first set of experiments used synthetic data to replicate a problem of automatic detection and classification of three types of ultrasonic pulses buried in noise (four-class classification). The second set of experiments analyzed two databases (one publicly available and one private) of real polysomnographic records from subjects with sleep disorders. These records were automatically staged in wake, rapid eye movement (REM) sleep, and non-REM sleep (three-class classification). Finally, the third set of experiments was performed on a publicly available database of single-channel real electroencephalographic data that included epileptic patients and healthy controls in five conditions (five-class classification). In all cases, alpha integration performed better than the considered single classifiers and classical fusion techniques.

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29. Information Geometry for Regularized Optimal Transport and Barycenters of Patterns

■■■-■■■ ■■■■■, *Ryo Karakida, Masafumi Oizumi, Marco Cuturi*

https://doi.org/10.1162/neco_a_01178 | pp. 827-848

We propose a new divergence on the manifold of probability distributions, building on the entropic regularization of optimal transportation problems. As Cuturi (2013) showed, regularizing the optimal transport problem with an entropic term is known to bring several computational benefits. However, because of that regularization, the resulting approximation of the optimal transport cost does not define a proper distance or divergence between probability distributions. We recently tried to introduce a family of divergences connecting the Wasserstein distance and the Kullback-Leibler divergence from an information geometry point of view (see Amari, Karakida, & Oizumi, 2018). However, that proposal was not able to retain key intuitive aspects of the Wasserstein geometry, such as translation invariance, which plays a key role when used in the more general problem of computing optimal transport barycenters. The divergence we propose in this work is able to retain such properties and admits an intuitive interpretation.

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30. Vector-Derived Transformation Binding: An Improved Binding Operation for Deep Symbol-Like Processing in Neural Networks

Jan Gosmann, Chris Eliasmith

https://doi.org/10.1162/neco_a_01179 | pp. 849-869

We present a new binding operation, vector-derived transformation binding (VTB), for use in vector symbolic architectures (VSA). The performance of VTB is compared to circular convolution, used in holographic reduced representations (HRRs), in terms of list and stack encoding capacity. A special focus is given to the possibility of a neural implementation by the means of the Neural Engineering Framework (NEF). While the scaling of required neural resources is slightly worse for VTB, it is found to be on par with circular convolution for list encoding and better for encoding of stacks. Furthermore, VTB influences the vector length less, which also benefits a neural implementation. Consequently, we argue that VTB is an

improvement over HRRs for neurally implemented VSAs.

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31. Inhibition and Excitation Shape Activity Selection: Effect of Oscillations in a Decision-Making Circuit

Thomas Bose, Andreagiovanni Reina, James A. R. Marshall

https://doi.org/10.1162/neco_a_01185 | pp. 870-896

Decision making is a complex task, and its underlying mechanisms that regulate behavior, such as the implementation of the coupling between physiological states and neural networks, are hard to decipher. To gain more insight into neural computations underlying ongoing binary decision-making tasks, we consider a neural circuit that guides the feeding behavior of a hypothetical animal making dietary choices. We adopt an inhibition motif from neural network theory and propose a dynamical system characterized by nonlinear feedback, which links mechanism (the implementation of the neural circuit and its coupling to the animal's nutritional state) and function (improving behavioral performance). A central inhibitory unit influences evidence-integrating excitatory units, which in our terms correspond to motivations competing for selection. We determine the parameter regime where the animal exhibits improved decision-making behavior and explain different behavioral outcomes by making the link between accessible states of the nonlinear neural circuit model and decision-making performance. We find that for given deficits in nutritional items, the variation of inhibition strength and ratio of excitation and inhibition strengths in the decision circuit allows the animal to enter an oscillatory phase that describes its internal motivational state. Our findings indicate that this oscillatory phase may improve the overall performance of the animal in an ongoing foraging task and underpin the importance of an integrated functional and mechanistic study of animal activity selection.

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32. Brain Morphometry Methods for Feature Extraction in Random Subspace Ensemble Neural Network Classification of First-Episode Schizophrenia

Roman Vyškovský, Daniel Schwarz, Tomáš Kašpárek

https://doi.org/10.1162/neco_a_01180 | pp. 897-918

Machine learning (ML) is a growing field that provides tools for automatic pattern recognition. The neuroimaging community currently tries to take advantage of ML in order to develop an auxiliary diagnostic tool for schizophrenia diagnostics. In this letter, we present a classification framework based on features extracted from magnetic resonance imaging (MRI) data using two automatic whole-brain morphometry methods: voxel-based (VBM) and deformation-based morphometry (DBM). The framework employs a random subspace ensemble-based artificial neural network classifier-in particular, a multilayer perceptron (MLP). The framework was tested on data from first-episode schizophrenia patients and healthy controls. The experiments differed in terms of feature extraction methods, using VBM, DBM, and a combination of both morphometry methods. Thus, features of different types were available for model adaptation. As we expected, the combination of features increased the MLP classification accuracy up to 73.12%-an improvement of 5% versus MLP-based only on VBM or DBM features. To further verify the findings, other comparisons using support vector machines in place of MLPs were made within the framework. However, it cannot be concluded that any classifier was better than another.

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33. Semisupervised Deep Stacking Network with Adaptive Learning Rate Strategy for Motor Imagery EEG Recognition

Xianlun Tang, Weichang Ma, Desong Kong, Wei Li

https://doi.org/10.1162/neco_a_01183 | pp. 919-942

Practical motor imagery electroencephalogram (EEG) data-based applications are limited by the waste of unlabeled samples in supervised learning and excessive time consumption in the pretraining period. A semisupervised deep stacking network with an adaptive learning rate strategy (SADSN) is proposed to solve the sample loss caused by supervised learning of EEG data and the extraction of manual features. The SADSN adopts the idea of an adaptive learning rate into a contrastive divergence (CD) algorithm to accelerate its convergence. Prior knowledge is introduced into the intermediary layer of the deep stacking network, and a restricted Boltzmann machine is trained by a semisupervised method in which the adjusting scope of the coefficient in learning rate is determined by performance analysis. Several EEG data sets are carried out to evaluate the performance of the proposed method. The results show that the recognition accuracy of SADSN is advanced with a more significant convergence rate and successfully classifies motor imagery.

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34. Multiple Timescale Online Learning Rules for Information Maximization with Energetic Constraints

Peng Yi, ShiNung Ching

https://doi.org/10.1162/neco_a_01182 | pp. 943-979

A key aspect of the neural coding problem is understanding how representations of afferent stimuli are built through the dynamics of learning and adaptation within neural networks. The infomax paradigm is built on the premise that such learning attempts to maximize the mutual information between input stimuli and neural activities. In this letter, we tackle the problem of such information-based neural coding with an eye toward two conceptual hurdles. Specifically, we examine and then show how this form of coding can be achieved with online input processing. Our framework thus obviates the biological incompatibility of optimization methods that rely on global network awareness and batch processing of sensory signals. Central to our result is the use of variational bounds as a surrogate objective function, an established technique that has not previously been shown to yield online policies. We obtain learning dynamics for both linear-continuous and discrete spiking neural encoding models under the umbrella of linear gaussian decoders. This result is enabled by approximating certain information quantities in terms of neuronal activity via pairwise feedback mechanisms. Furthermore, we tackle the problem of how such learning dynamics can be realized with strict energetic constraints. We show that endowing networks with auxiliary variables that evolve on a slower timescale can allow for the realization of saddle-point optimization within the neural dynamics, leading to neural codes with favorable properties in terms of both information and energy.

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35. Introducing User-Prescribed Constraints in Markov Chains for Nonlinear Dimensionality Reduction

Purushottam D. Dixit

https://doi.org/10.1162/neco_a_01184 | pp. 980-997

Stochastic kernel-based dimensionality-reduction approaches have become popular in the past decade. The central component of many of these methods is a symmetric kernel that quantifies the vicinity between pairs of data points and a kernel-induced Markov chain on the data. Typically, the Markov chain is fully specified by the kernel through row normalization. However, in many cases, it is desirable to impose user-specified stationary-state and dynamical constraints on the Markov chain. Unfortunately, no systematic framework exists to impose such user-defined constraints. Here, based on our previous work on inference of Markov models, we introduce a path entropy maximization based approach to derive the transition probabilities of Markov chains using a kernel and additional user-specified constraints. We illustrate the usefulness of these Markov chains with examples.

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36. Sparse Associative Memory

H. Hoffmann

https://doi.org/10.1162/neco_a_01181 | pp. 998-1014

It is still unknown how associative biological memories operate. Hopfield networks are popular models of associative memory, but they suffer from spurious memories and low efficiency. Here, we present a new model of an associative memory that overcomes these deficiencies. We call this model sparse associative memory (SAM) because it is based on sparse projections from neural patterns to pattern-specific neurons. These sparse projections have been shown to be sufficient to uniquely encode a neural pattern. Based on this principle, we investigate theoretically and in simulation our SAM model, which turns out to have high memory efficiency and a vanishingly small probability of spurious memories. This model may serve as a basic building block of brain functions involving associative memory.

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37. Quantifying Information Conveyed by Large Neuronal Populations

John Berkowitz, Tatyana O. Sharpee

https://doi.org/10.1162/neco_a_01193 | pp. 1015-1047

Quantifying mutual information between inputs and outputs of a large neural circuit is an important open problem in both machine learning and neuroscience. However, evaluation of the mutual information is known to be generally intractable for large systems due to the exponential growth in the number of terms that need to be evaluated. Here we show how information contained in the responses of large neural populations can be effectively computed provided the input-output functions of individual neurons can be measured and approximated by a logistic function applied to a potentially nonlinear function of the stimulus. Neural responses in this model can remain sensitive to multiple stimulus components. We show that the mutual information in this model can be effectively approximated as a sum of lower-dimensional conditional mutual information terms. The approximations become exact in the limit of large neural populations and for certain conditions on the distribution of receptive fields across the neural population. We empirically find that these approximations continue to work well even when the conditions on the receptive field distributions are not fulfilled. The computing cost for the proposed methods grows linearly in the dimension of the input and compares favorably with other approximations.

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38. Controlling Complexity of Cerebral Cortex Simulations—II: Streamlined Microcircuits

Henri Hokkanen, Vafa Andalibi, Simo Vanni

https://doi.org/10.1162/neco_a_01188 | pp. 1066-1084

Recently, Markram et al. (2015) presented a model of the rat somatosensory microcircuit (Markram model). Their model is high in anatomical and physiological detail, and its simulation requires supercomputers. The lack of neuroinformatics and computing power is an obstacle for using a similar approach to build models of other cortical areas or larger cortical systems. Simplified neuron models offer an attractive alternative to high-fidelity Hodgkin-Huxley-type neuron models, but their validity in modeling cortical circuits is unclear. We simplified the Markram model to a network of exponential integrate-and-fire (EIF) neurons that runs on a single CPU core in reasonable time. We analyzed the electrophysiology and the morphology of the Markram model neurons with eFel and NeuroM tools, provided by the Blue Brain Project. We then constructed neurons with few compartments and averaged parameters from the reference model. We used the CxSystem simulation framework to explore the role of short-term plasticity and GABA[Formula: see text] and NMDA synaptic conductances in replicating oscillatory phenomena in the Markram model. We show that having a slow inhibitory synaptic conductance (GABA[Formula: see text] allows replication of oscillatory behavior in the high-calcium state. Furthermore, we show that qualitatively similar dynamics are seen even with a reduced number of cell types (from 55 to 17 types). This reduction halved the computation time. Our results suggest that qualitative dynamics of cortical microcircuits can be studied using limited neuroinformatics and computing resources supporting parameter exploration and simulation of cortical systems. The simplification procedure can easily be adapted to studying other microcircuits for which sparse electrophysiological and morphological data are available.

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39. Decoding Movements from Cortical Ensemble Activity Using a Long Short-Term Memory Recurrent Network

Po-He Tseng, Núria Armengol Urpí, Mikhail Lebedev, Miguel A. L. Nicolelis

https://doi.org/10.1162/neco_a_01189 | pp. 1085-1113

Although many real-time neural decoding algorithms have been proposed for brain-machine interface (BMI) applications over the years, an optimal, consensual approach remains elusive. Recent advances in deep learning algorithms provide new opportunities for improving the design of BMI decoders, including the use of recurrent artificial neural networks to decode neuronal ensemble activity in real time. Here, we developed a long-short term memory (LSTM) decoder for extracting movement kinematics from the activity of large (N = 134–402) populations of neurons, sampled simultaneously from multiple cortical areas, in rhesus monkeys performing motor tasks. Recorded regions included primary motor, dorsal premotor, supplementary motor, and primary somatosensory cortical areas. The LSTM's capacity to retain information for extended periods of time enabled accurate decoding for tasks that required both movements and periods of immobility. Our LSTM algorithm significantly outperformed the state-of-the-art unscented Kalman filter when applied to three tasks: center-out arm reaching, bimanual reaching, and bipedal walking on a treadmill. Notably, LSTM units exhibited a variety of well-known physiological features of cortical neuronal activity, such as directional tuning and neuronal dynamics across task epochs. LSTM modeled several key physiological attributes of cortical circuits involved in motor tasks. These findings suggest that LSTM-based approaches could yield a better algorithm strategy for neuroprostheses that employ BMIs to restore movement in severely disabled

patients.

40. Asynchronous Event-Based Motion Processing: From Visual Events to Probabilistic Sensory Representation

Mina A. Khoei, Sio-Hoï Ieng, Ryad Benosman

https://doi.org/10.1162/neco_a_01191 | pp. 1114-1138

In this work, we propose a two-layered descriptive model for motion processing from retina to the cortex, with an event-based input from the asynchronous time-based image sensor (ATIS) camera. Spatial and spatiotemporal filtering of visual scenes by motion energy detectors has been implemented in two steps in a simple layer of a lateral geniculate nucleus model and a set of three-dimensional Gabor kernels, eventually forming a probabilistic population response. The high temporal resolution of independent and asynchronous local sensory pixels from the ATIS provides a realistic stimulation to study biological motion processing, as well as developing bio-inspired motion processors for computer vision applications. Our study combines two significant theories in neuroscience: event-based stimulation and probabilistic sensory representation. We have modeled how this might be done at the vision level, as well as suggesting this framework as a generic computational principle among different sensory modalities.

41. A Geometrical Analysis of Global Stability in Trained Feedback Networks

Francesca Mastrogiuseppe, Srdjan Ostojic

https://doi.org/10.1162/neco_a_01187 | pp. 1139-1182

Recurrent neural networks have been extensively studied in the context of neuroscience and machine learning due to their ability to implement complex computations. While substantial progress in designing effective learning algorithms has been achieved, a full understanding of trained recurrent networks is still lacking. Specifically, the mechanisms that allow computations to emerge from the underlying recurrent dynamics are largely unknown. Here we focus on a simple yet underexplored computational setup: a feedback architecture trained to associate a stationary output to a stationary input. As a starting point, we derive an approximate analytical description of global dynamics in trained networks, which assumes uncorrelated connectivity weights in the feedback and in the random bulk. The resulting mean-field theory suggests that the task admits several classes of solutions, which imply different stability properties. Different classes are characterized in terms of the geometrical arrangement of the readout with respect to the input vectors, defined in the high-dimensional space spanned by the network population. We find that such an approximate theoretical approach can be used to understand how standard training techniques implement the input-output task in finite-size feedback networks. In particular, our simplified description captures the local and the global stability properties of the target solution, and thus predicts training performance.

42. Learning Moral Graphs in Construction of High-Dimensional Bayesian Networks for Mixed Data

Suwa Xu, Bochao Jia, Faming Liang

https://doi.org/10.1162/neco_a_01190 | pp. 1183-1214

Bayesian networks have been widely used in many scientific fields for describing the conditional independence relationships for a large set of random variables. This letter proposes a novel algorithm, the so-called p-learning algorithm, for learning moral graphs for high-dimensional Bayesian networks. The moral graph is a Markov network representation of the Bayesian network and also the key to construction of the Bayesian network for constraint-based algorithms. The consistency of the p-learning algorithm is justified under the small- n, large- p scenario. The numerical results indicate that the p-learning algorithm significantly outperforms the existing ones, such as the PC, grow-shrink, incremental association, semi-interleaved hiton, hill-climbing, and max-min hill-climbing. Under the sparsity assumption, the p-learning algorithm has a computational complexity of O(p 2) even in the worst case, while the existing algorithms have a computational complexity of O(p 3) in the worst case.

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43. Improving the Antinoise Ability of DNNs via a Bio-Inspired Noise Adaptive Activation Function Rand Softplus

Yunhua Chen, Yingchao Mai, Jinsheng Xiao, Ling Zhang

https://doi.org/10.1162/neco_a_01192 | pp. 1215-1233

Although deep neural networks (DNNs) have led to many remarkable results in cognitive tasks, they are still far from catching up with human-level cognition in antinoise capability. New research indicates how brittle and susceptible current models are to small variations in data distribution. In this letter, we study the stochasticity-resistance character of biological neurons by simulating the input-output response process of a leaky integrate-and-fire (LIF) neuron model and proposed a novel activation function, rand softplus (RSP), to model the response process. In RSP, a scale factor [Formula: see text] is employed to mimic the stochasticity-adaptability of biological neurons, thereby enabling the antinoise capability of a DNN to be improved by the novel activation function. We validated the performance of RSP with a 19-layer residual network (ResNet) and a 19-layer visual geometry group (VGG) on facial expression recognition data sets and compared it with other popular activation functions, such as rectified linear units (ReLU), softplus, leaky ReLU (LReLU), exponential linear unit (ELU), and noisy softplus (NSP). The experimental results show that RSP is applied to VGG-19 or ResNet-19, and the average recognition accuracy under five different noise levels exceeds the other functions on both of the two facial expression data sets; in other words, RSP outperforms the other activation functions in noise resistance. Compared with the application in ResNet-19, the application of RSP in VGG-19 can improve a network's antinoise performance to a greater extent. In addition, RSP is easier to train compared to NSP because it has only one parameter to be calculated automatically according to the input data. Therefore, this work provides the deep learning community with a novel activation function that can better deal with overfitting problems.

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44. A Review of Recurrent Neural Networks: LSTM Cells and Network Architectures

Yong Yu, Xiaosheng Si, Changhua Hu, Jianxun Zhang

https://doi.org/10.1162/neco_a_01199 | pp. 1235-1270

Recurrent neural networks (RNNs) have been widely adopted in research areas concerned with sequential data, such as text, audio, and video. However, RNNs consisting of sigma cells or tanh cells are unable to learn the relevant information of input data when the input gap is large. By introducing gate functions into the

cell structure, the long short-term memory (LSTM) could handle the problem of long-term dependencies well. Since its introduction, almost all the exciting results based on RNNs have been achieved by the LSTM. The LSTM has become the focus of deep learning. We review the LSTM cell and its variants to explore the learning capacity of the LSTM cell. Furthermore, the LSTM networks are divided into two broad categories: LSTM-dominated networks and integrated LSTM networks. In addition, their various applications are discussed. Finally, future research directions are presented for LSTM networks.

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45. Characterizing Brain Connectivity From Human Electrocorticography Recordings With Unobserved Inputs During Epileptic Seizures

Anup Das, Daniel Sexton, Claudia Lainscsek, Sydney S. Cash, Terrence J. Sejnowski

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Epilepsy is a neurological disorder characterized by the sudden occurrence of unprovoked seizures. There is extensive evidence of significantly altered brain connectivity during seizure periods in the human brain. Research on analyzing human brain functional connectivity during epileptic seizures has been limited predominantly to the use of the correlation method. However, spurious connectivity can be measured between two brain regions without having direct connection or interaction between them. Correlations can be due to the apparent interactions of the two brain regions resulting from common input from a third region, which may or may not be observed. Hence, researchers have recently proposed a sparse-plus-latent-regularized precision matrix (SLRPM) when there are unobserved or latent regions interacting with the observed regions. The SLRPM method yields partial correlations of the conditional statistics of the observed regions given the latent regions, thus identifying observed regions that are conditionally independent of both the observed and latent regions. We evaluate the performance of the methods using a spring-mass artificial network and assuming that some nodes cannot be observed, thus constituting the latent variables in the example. Several cases have been considered, including both sparse and dense connections, short-range and long-range connections, and a varying number of latent variables. The SLRPM method is then applied to estimate brain connectivity during epileptic seizures from human ECoG recordings. Seventy-four clinical seizures from five patients, all having complex partial epilepsy, were analyzed using SLRPM, and brain connectivity was quantified using modularity index, clustering coefficient, and eigenvector centrality. Furthermore, using a measure of latent inputs estimated by the SLRPM method, it was possible to automatically detect 72 of the 74 seizures with four false positives and find six seizures that were not marked manually.

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46. Multi-Input, Multi-Output Neuronal Mode Network Approach to Modeling the Encoding Dynamics and Functional Connectivity of Neural Systems

Kunling Geng, Dae C. Shin, Dong Song, Robert E. Hampson, Samuel A. Deadwyler, Theodore W. Berger, Vasilis Z. Marmarelis

https://doi.org/10.1162/neco_a_01204 | pp. 1327-1355

This letter proposes a novel method, multi-input, multi-output neuronal mode network (MIMO-NMN), for modeling encoding dynamics and functional connectivity in neural ensembles such as the hippocampus. Compared with conventional approaches such as the Volterra-Wiener model, linear-nonlinear-cascade (LNC) model, and generalized linear model (GLM), the NMN has several advantages in terms of estimation

accuracy, model interpretation, and functional connectivity analysis. We point out the limitations of current neural spike modeling methods, especially the estimation biases caused by the imbalanced class problem when the number of zeros is significantly larger than ones in the spike data. We use synthetic data to test the performance of NMN with a comparison of the traditional methods, and the results indicate the NMN approach could reduce the imbalanced class problem and achieve better predictions. Subsequently, we apply the MIMO-NMN method to analyze data from the human hippocampus. The results indicate that the MIMO-NMN method is a promising approach to modeling neural dynamics and analyzing functional connectivity of multi-neuronal data.

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47. Effective Dimensionality Reduction for Visualizing Neural Dynamics by Laplacian Eigenmaps

Guanghao Sun, Shaomin Zhang, Yiwei Zhang, Kedi Xu, Qiaosheng Zhang, Ting Zhao, Xiaoxiang Zheng
https://doi.org/10.1162/neco_a_01203 | pp. 1356-1379

With the development of neural recording technology, it has become possible to collect activities from hundreds or even thousands of neurons simultaneously. Visualization of neural population dynamics can help neuroscientists analyze large-scale neural activities efficiently. In this letter, Laplacian eigenmaps is applied to this task for the first time, and the experimental results show that the proposed method significantly outperforms the commonly used methods. This finding was confirmed by the systematic evaluation using nonhuman primate data, which contained the complex dynamics well suited for testing. According to our results, Laplacian eigenmaps is better than the other methods in various ways and can clearly visualize interesting biological phenomena related to neural dynamics.

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48. A Computational Perspective of the Role of the Thalamus in Cognition

Nima Dehghani, Ralf Wimmer
https://doi.org/10.1162/neco_a_01197 | pp. 1380-1418

The thalamus has traditionally been considered as only a relay source of cortical inputs, with hierarchically organized cortical circuits serially transforming thalamic signals to cognitively relevant representations. Given the absence of local excitatory connections within the thalamus, the notion of thalamic relay seemed like a reasonable description over the past several decades. Recent advances in experimental approaches and theory provide a broader perspective on the role of the thalamus in cognitively relevant cortical computations and suggest that only a subset of thalamic circuit motifs fits the relay description. Here, we discuss this perspective and highlight the potential role for the thalamus, and specifically the mediodorsal (MD) nucleus, in the dynamic selection of cortical representations through a combination of intrinsic thalamic computations and output signals that change cortical network functional parameters. We suggest that through the contextual modulation of cortical computation, the thalamus and cortex jointly optimize the information and cost trade-off in an emergent fashion. We emphasize that coordinated experimental and theoretical efforts will provide a path to understanding the role of the thalamus in cognition, along with an understanding to augment cognitive capacity in health and disease.

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49. A Case of Near-Optimal Sensory Integration Based on Kohonen Self-Organizing Maps

Lennart Gustafsson

https://doi.org/10.1162/neco_a_01200 | pp. 1419-1429

This letter shows by digital simulation that a simple rule applied to one-dimensional self-organized maps for integrating sensory perceptions from two identical sources yielding position information as integers, corrupted by independent noise sources, yields almost statistically optimal results for position estimation as determined by maximum likelihood estimation. There is no learning of the corrupting noise sources nor is any information about the statistics of the noise sources available to the integrating process. The simple rule employed yields a measure of the quality of the estimated position of the source. The letter also shows that if the Bayesian estimates, which are rational numbers, are rounded in order to comply with the stipulation that integers be identified, the Bayesian estimation will have a larger variance than the proposed integration.

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50. A Reservoir Computing Model of Reward-Modulated Motor Learning and Automaticity

Ryan Pyle, Robert Rosenbaum

https://doi.org/10.1162/neco_a_01198 | pp. 1430-1461

Reservoir computing is a biologically inspired class of learning algorithms in which the intrinsic dynamics of a recurrent neural network are mined to produce target time series. Most existing reservoir computing algorithms rely on fully supervised learning rules, which require access to an exact copy of the target response, greatly reducing the utility of the system. Reinforcement learning rules have been developed for reservoir computing, but we find that they fail to converge on complex motor tasks. Current theories of biological motor learning pose that early learning is controlled by dopamine-modulated plasticity in the basal ganglia that trains parallel cortical pathways through unsupervised plasticity as a motor task becomes well learned. We developed a novel learning algorithm for reservoir computing that models the interaction between reinforcement and unsupervised learning observed in experiments. This novel learning algorithm converges on simulated motor tasks on which previous reservoir computing algorithms fail and reproduces experimental findings that relate Parkinson's disease and its treatments to motor learning. Hence, incorporating biological theories of motor learning improves the effectiveness and biological relevance of reservoir computing models.

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51. Effect of Depth and Width on Local Minima in Deep Learning

Kenji Kawaguchi, Jiaoyang Huang, Leslie Pack Kaelbling

https://doi.org/10.1162/neco_a_01195 | pp. 1462-1498

In this paper, we analyze the effects of depth and width on the quality of local minima, without strong overparameterization and simplification assumptions in the literature. Without any simplification assumption, for deep nonlinear neural networks with the squared loss, we theoretically show that the quality of local minima tends to improve toward the global minimum value as depth and width increase. Furthermore, with a locally induced structure on deep nonlinear neural networks, the values of local minima of neural networks are theoretically proven to be no worse than the globally optimal values of corresponding classical machine

learning models. We empirically support our theoretical observation with a synthetic data set, as well as MNIST, CIFAR-10, and SVHN data sets. When compared to previous studies with strong overparameterization assumptions, the results in this letter do not require overparameterization and instead show the gradual effects of overparameterization as consequences of general results.

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52. Matrix Product State–Based Quantum Classifier

Amandeep Singh Bhatia, Mandeep Kaur Saggi, Ajay Kumar, Sushma Jain

https://doi.org/10.1162/neco_a_01202 | pp. 1499-1517

Interest in quantum computing has increased significantly. Tensor network theory has become increasingly popular and widely used to simulate strongly entangled correlated systems. Matrix product state (MPS) is a well-designed class of tensor network states that plays an important role in processing quantum information. In this letter, we show that MPS, as a one-dimensional array of tensors, can be used to classify classical and quantum data. We have performed binary classification of the classical machine learning data set Iris encoded in a quantum state. We have also investigated its performance by considering different parameters on the ibmqx4 quantum computer and proved that MPS circuits can be used to attain better accuracy. Furthermore the learning ability of an MPS quantum classifier is tested to classify evapotranspiration (ET[Formula: see text]) for the Patiala meteorological station located in northern Punjab (India), using three years of a historical data set (Agri). We have used different performance metrics of classification to measure its capability. Finally, the results are plotted and the degree of correspondence among values of each sample is shown.

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53. On the Organization of Grid and Place Cells: Neural Denoising via Subspace Learning

David Schwartz, O. Ozan Koyluoglu

https://doi.org/10.1162/neco_a_01208 | pp. 1519-1550

Place cells in the hippocampus (HC) are active when an animal visits a certain location (referred to as a place field) within an environment. Grid cells in the medial entorhinal cortex (MEC) respond at multiple locations, with firing fields that form a periodic and hexagonal tiling of the environment. The joint activity of grid and place cell populations, as a function of location, forms a neural code for space. In this article, we develop an understanding of the relationships between coding theoretically relevant properties of the combined activity of these populations and how these properties limit the robustness of this representation to noise-induced interference. These relationships are revisited by measuring the performances of biologically realizable algorithms implemented by networks of place and grid cell populations, as well as constraint neurons, which perform denoising operations. Contributions of this work include the investigation of coding theoretic limitations of the mammalian neural code for location and how communication between grid and place cell networks may improve the accuracy of each population's representation. Simulations demonstrate that denoising mechanisms analyzed here can significantly improve the fidelity of this neural representation of space. Furthermore, patterns observed in connectivity of each population of simulated cells predict that anti-Hebbian learning drives decreases in inter-HC-MEC connectivity along the dorsoventral axis.

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54. Approximating the Architecture of Visual Cortex in a Convolutional Network

Bryan Tripp

https://doi.org/10.1162/neco_a_01211 | pp. 1551-1591

Deep convolutional neural networks (CNNs) have certain structural, mechanistic, representational, and functional parallels with primate visual cortex and also many differences. However, perhaps some of the differences can be reconciled. This study develops a cortex-like CNN architecture, via (1) a loss function that quantifies the consistency of a CNN architecture with neural data from tract tracing, cell reconstruction, and electrophysiology studies; (2) a hyperparameter-optimization approach for reducing this loss, and (3) heuristics for organizing units into convolutional-layer grids. The optimized hyperparameters are consistent with neural data. The cortex-like architecture differs from typical CNN architectures. In particular, it has longer skip connections, larger kernels and strides, and qualitatively different connection sparsity. Importantly, layers of the cortex-like network have one-to-one correspondences with cortical neuron populations. This should allow unambiguous comparison of model and brain representations in the future and, consequently, more precise measurement of progress toward more biologically realistic deep networks.

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55. On the Complexity of Logistic Regression Models

Nicola Bulso, Matteo Marsili, Yasser Roudi

https://doi.org/10.1162/neco_a_01207 | pp. 1592-1623

We investigate the complexity of logistic regression models, which is defined by counting the number of indistinguishable distributions that the model can represent (Balasubramanian, 1997). We find that the complexity of logistic models with binary inputs depends not only on the number of parameters but also on the distribution of inputs in a nontrivial way that standard treatments of complexity do not address. In particular, we observe that correlations among inputs induce effective dependencies among parameters, thus constraining the model and, consequently, reducing its complexity. We derive simple relations for the upper and lower bounds of the complexity. Furthermore, we show analytically that defining the model parameters on a finite support rather than the entire axis decreases the complexity in a manner that critically depends on the size of the domain. Based on our findings, we propose a novel model selection criterion that takes into account the entropy of the input distribution. We test our proposal on the problem of selecting the input variables of a logistic regression model in a Bayesian model selection framework. In our numerical tests, we find that while the reconstruction errors of standard model selection approaches (AIC, BIC, [Formula: see text] regularization) strongly depend on the sparsity of the ground truth, the reconstruction error of our method is always close to the minimum in all conditions of sparsity, data size, and strength of input correlations. Finally, we observe that when considering categorical instead of binary inputs, in a simple and mathematically tractable case, the contribution of the alphabet size to the complexity is very small compared to that of parameter space dimension. We further explore the issue by analyzing the data set of the “13 keys to the White House,” a method for forecasting the outcomes of US presidential elections.

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56. When Not to Classify: Anomaly Detection of Attacks (ADA) on DNN Classifiers at Test Time

David J. Miller, Yujia Wang, George Kesidis
https://doi.org/10.1162/neco_a_01209 | pp. 1624-1670

A significant threat to the recent, wide deployment of machine learning-based systems, including deep neural networks (DNNs), is adversarial learning attacks. The main focus here is on evasion attacks against DNN-based classifiers at test time. While much work has focused on devising attacks that make small perturbations to a test pattern (e.g., an image) that induce a change in the classifier's decision, until recently there has been a relative paucity of work defending against such attacks. Some works robustify the classifier to make correct decisions on perturbed patterns. This is an important objective for some applications and for natural adversary scenarios. However, we analyze the possible digital evasion attack mechanisms and show that in some important cases, when the pattern (image) has been attacked, correctly classifying it has no utility---when the image to be attacked is (even arbitrarily) selected from the attacker's cache and when the sole recipient of the classifier's decision is the attacker. Moreover, in some application domains and scenarios, it is highly actionable to detect the attack irrespective of correctly classifying in the face of it (with classification still performed if no attack is detected). We hypothesize that adversarial perturbations are machine detectable even if they are small. We propose a purely unsupervised anomaly detector (AD) that, unlike previous works, (1) models the joint density of a deep layer using highly suitable null hypothesis density models (matched in particular to the nonnegative support for rectified linear unit (ReLU) layers); (2) exploits multiple DNN layers; and (3) leverages a source and destination class concept, source class uncertainty, the class confusion matrix, and DNN weight information in constructing a novel decision statistic grounded in the Kullback-Leibler divergence. Tested on MNIST and CIFAR image databases under three prominent attack strategies, our approach outperforms previous detection methods, achieving strong receiver operating characteristic area under the curve detection accuracy on two attacks and better accuracy than recently reported for a variety of methods on the strongest (CW) attack. We also evaluate a fully white box attack on our system and demonstrate that our method can be leveraged to strong effect in detecting reverse engineering attacks. Finally, we evaluate other important performance measures such as classification accuracy versus true detection rate and multiple measures versus attack strength.

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57. Learning Compositional Representations of Interacting Systems with Restricted Boltzmann Machines: Comparative Study of Lattice Proteins

Jérôme Tubiana, Simona Cocco, Rémi Monasson
https://doi.org/10.1162/neco_a_01210 | pp. 1671-1717

A restricted Boltzmann machine (RBM) is an unsupervised machine learning bipartite graphical model that jointly learns a probability distribution over data and extracts their relevant statistical features. RBMs were recently proposed for characterizing the patterns of coevolution between amino acids in protein sequences and for designing new sequences. Here, we study how the nature of the features learned by RBM changes with its defining parameters, such as the dimensionality of the representations (size of the hidden layer) and the sparsity of the features. We show that for adequate values of these parameters, RBMs operate in a so-called compositional phase in which visible configurations sampled from the RBM are obtained by recombining these features. We then compare the performance of RBM with other standard representation learning algorithms, including principal or independent component analysis (PCA, ICA), autoencoders (AE), variational autoencoders (VAE), and their sparse variants. We show that RBMs, due to the stochastic mapping between data configurations and representations, better capture the underlying interactions in the

system and are significantly more robust with respect to sample size than deterministic methods such as PCA or ICA. In addition, this stochastic mapping is not prescribed a priori as in VAE, but learned from data, which allows RBMs to show good performance even with shallow architectures. All numerical results are illustrated on synthetic lattice protein data that share similar statistical features with real protein sequences and for which ground-truth interactions are known.

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58. Variable Selection for Nonparametric Learning with Power Series Kernels

Kota Matsui, Wataru Kumagai, Kenta Kanamori, Mitsuaki Nishikimi, Takafumi Kanamori

https://doi.org/10.1162/neco_a_01212 | pp. 1718-1750

In this letter, we propose a variable selection method for general nonparametric kernel-based estimation. The proposed method consists of two-stage estimation: (1) construct a consistent estimator of the target function, and (2) approximate the estimator using a few variables by [Formula: see text]-type penalized estimation. We see that the proposed method can be applied to various kernel nonparametric estimation such as kernel ridge regression, kernel-based density, and density-ratio estimation. We prove that the proposed method has the property of variable selection consistency when the power series kernel is used. Here, the power series kernel is a certain class of kernels containing polynomial and exponential kernels. This result is regarded as an extension of the variable selection consistency for the nonnegative garrote (NNG), a special case of the adaptive Lasso, to the kernel-based estimators. Several experiments, including simulation studies and real data applications, show the effectiveness of the proposed method.

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59. Decoding Hidden Cognitive States From Behavior and Physiology Using a Bayesian Approach

Ali Yousefi, Ishita Basu, Angelique C. Paulk, Noam Peled, Emad N. Eskandar, Darin D. Dougherty, Sydney S. Cash, Alik S. Widge, Uri T. Eden

https://doi.org/10.1162/neco_a_01196 | pp. 1751-1788

Cognitive processes, such as learning and cognitive flexibility, are both difficult to measure and to sample continuously using objective tools because cognitive processes arise from distributed, high-dimensional neural activity. For both research and clinical applications, that dimensionality must be reduced. To reduce dimensionality and measure underlying cognitive processes, we propose a modeling framework in which a cognitive process is defined as a low-dimensional dynamical latent variable—called a cognitive state, which links high-dimensional neural recordings and multidimensional behavioral readouts. This framework allows us to decompose the hard problem of modeling the relationship between neural and behavioral data into separable encoding-decoding approaches. We first use a state-space modeling framework, the behavioral decoder, to articulate the relationship between an objective behavioral readout (e.g., response times) and cognitive state. The second step, the neural encoder, involves using a generalized linear model (GLM) to identify the relationship between the cognitive state and neural signals, such as local field potential (LFP). We then use the neural encoder model and a Bayesian filter to estimate cognitive state using neural data (LFP power) to generate the neural decoder. We provide goodness-of-fit analysis and model selection criteria in support of the encoding-decoding result. We apply this framework to estimate an underlying cognitive state from neural data in human participants ([Formula: see text]) performing a cognitive conflict task. We successfully estimated the cognitive state within the 95% confidence intervals of that estimated

using behavior readout for an average of 90% of task trials across participants. In contrast to previous encoder-decoder models, our proposed modeling framework incorporates LFP spectral power to encode and decode a cognitive state. The framework allowed us to capture the temporal evolution of the underlying cognitive processes, which could be key to the development of closed-loop experiments and treatments.

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60. Sensitivity to Stimulus Irregularity Is Inherent in Neural Networks

Teun van Gils, Paul Tiesinga, Bernhard Englitz, Marijn Bart Martens

https://doi.org/10.1162/neco_a_01215 | pp. 1789-1824

Behavior is controlled by complex neural networks in which neurons process thousands of inputs. However, even short spike trains evoked in a single cortical neuron were demonstrated to be sufficient to influence behavior in vivo. Specifically, irregular sequences of interspike intervals (ISIs) had a more reliable influence on behavior despite their resemblance to stochastic activity. Similarly, irregular tactile stimulation led to higher rates of behavioral responses. In this study, we identify the mechanisms enabling this sensitivity to stimulus irregularity (SSI) on the neuronal and network levels using simulated spiking neural networks. Matching in vivo experiments, we find that irregular stimulation elicits more detectable network events (bursts) than regular stimulation. Dissecting the stimuli, we identify short ISIs-occurring more frequently in irregular stimulations-as the main drivers of SSI rather than complex irregularity per se. In addition, we find that short-term plasticity modulates SSI. We subsequently eliminate the different mechanisms in turn to assess their role in generating SSI. Removing inhibitory interneurons, we find that SSI is retained, suggesting that SSI is not dependent on inhibition. Removing recurrency, we find that SSI is retained due to the ability of individual neurons to integrate activity over short timescales ("cell memory"). Removing single-neuron dynamics, we find that SSI is retained based on the short-term retention of activity within the recurrent network structure ("network memory"). Finally, using a further simplified probabilistic model, we find that local network structure is not required for SSI. Hence, SSI is identified as a general property that we hypothesize to be ubiquitous in neural networks with different structures and biophysical properties. Irregular sequences contain shorter ISIs, which are the main drivers underlying SSI. The experimentally observed SSI should thus generalize to other systems, suggesting a functional role for irregular activity in cortex.

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61. Learning with Precise Spike Times: A New Decoding Algorithm for Liquid State Machines

Dorian Florescu, Daniel Coca

https://doi.org/10.1162/neco_a_01218 | pp. 1825-1852

There is extensive evidence that biological neural networks encode information in the precise timing of the spikes generated and transmitted by neurons, which offers several advantages over rate-based codes. Here we adopt a vector space formulation of spike train sequences and introduce a new liquid state machine (LSM) network architecture and a new forward orthogonal regression algorithm to learn an input-output signal mapping or to decode the brain activity. The proposed algorithm uses precise spike timing to select the presynaptic neurons relevant to each learning task. We show that using precise spike timing to train the LSM and selecting the readout presynaptic neurons leads to a significant increase in performance on binary classification tasks, in decoding neural activity from multielectrode array recordings, as well as in a speech recognition task, compared with what is achieved using the standard architecture and training methods.

62. A New Nonlinear Sparse Component Analysis for a Biologically Plausible Model of Neurons

S. M. Heidarieh, Mehran Jahed, Ali Ghazizadeh

https://doi.org/10.1162/neco_a_01214 | pp. 1853-1873

It is known that brain can create a sparse representation of the environment in both sensory and mnemonic forms (Olshausen & Field, 2004). Such sparse representation can be combined in downstream areas to create rich multisensory responses to support various cognitive and motor functions. Determining the components present in neuronal responses in a given region is key to deciphering its functional role and connection with upstream areas. One approach for parsing out various sources of information in a single neuron is by using linear blind source separation (BSS) techniques. However, applying linear techniques to neuronal spiking activity is likely to be suboptimal due to inherent and unknown nonlinearity of neuronal responses to inputs. This letter proposes a nonlinear sparse component analysis (SCA) method to separate jointly sparse inputs to neurons with post summation nonlinearity, or SCA for post-nonlinear neurons (SCAPL). Specifically, a linear clustering approach followed by principal curve regression (PCR) and a nonlinear curve fitting are used to separate sources. Analysis using simulated data shows that SCAPL accuracy outperforms ones obtained by linear SCA, as well as other separating methods, including linear independent and principal component analyses. In SCAPL, the number of derived sparse components is not limited by the number of neurons, unlike most BSS methods. Furthermore, this method allows for a broad range of post-summation nonlinearities that could differ among neurons. The sensitivity of our method to noise, joint sparseness, degree, and shape of nonlinearity and mixing ill conditions is discussed and compared to existing methods. Our results show that the proposed method can successfully separate input components in a population of neurons provided that they are temporally sparse to some degree. Application of SCAPL should facilitate comparison of functional roles across regions by parsing various elements present in a region.

63. Hidden Markov Models Predict the Future Choice Better Than a PSTH-Based Method

Encarni Marcos, Fabrizio Londei, Aldo Genovesio

https://doi.org/10.1162/neco_a_01216 | pp. 1874-1890

Beyond average firing rate, other measurable signals of neuronal activity are fundamental to an understanding of behavior. Recently, hidden Markov models (HMMs) have been applied to neural recordings and have described how neuronal ensembles process information by going through sequences of different states. Such collective dynamics are impossible to capture by just looking at the average firing rate. To estimate how well HMMs can decode information contained in single trials, we compared HMMs with a recently developed classification method based on the peristimulus time histogram (PSTH). The accuracy of the two methods was tested by using the activity of prefrontal neurons recorded while two monkeys were engaged in a strategy task. In this task, the monkeys had to select one of three spatial targets based on an instruction cue and on their previous choice. We show that by using the single trial's neural activity in a period preceding action execution, both models were able to classify the monkeys' choice with an accuracy higher than by chance. Moreover, the HMM was significantly more accurate than the PSTH-based method, even in cases in which the HMM performance was low, although always above

chance. Furthermore, the accuracy of both methods was related to the number of neurons exhibiting spatial selectivity within an experimental session. Overall, our study shows that neural activity is better described when not only the mean activity of individual neurons is considered and that therefore, the study of other signals rather than only the average firing rate is fundamental to an understanding of the dynamics of neuronal ensembles.

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64. Supervised Determined Source Separation with Multichannel Variational Autoencoder

Hirokazu Kameoka, Li Li, Shota Inoue, Shoji Makino

https://doi.org/10.1162/neco_a_01217 | pp. 1891-1914

This letter proposes a multichannel source separation technique, the multichannel variational autoencoder (MVAE) method, which uses a conditional VAE (CVAE) to model and estimate the power spectrograms of the sources in a mixture. By training the CVAE using the spectrograms of training examples with source-class labels, we can use the trained decoder distribution as a universal generative model capable of generating spectrograms conditioned on a specified class index. By treating the latent space variables and the class index as the unknown parameters of this generative model, we can develop a convergence-guaranteed algorithm for supervised determined source separation that consists of iteratively estimating the power spectrograms of the underlying sources, as well as the separation matrices. In experimental evaluations, our MVAE produced better separation performance than a baseline method.

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65. A Real-Time Health 4.0 Framework with Novel Feature Extraction and Classification for Brain-Controlled IoT-Enabled Environments

B. Jagadish, Pranav Mishra, M. P. R. S. Kiran, P. Rajalakshmi

https://doi.org/10.1162/neco_a_01223 | pp. 1915-1944

In this letter, we propose two novel methods for four-class motor imagery (MI) classification using electroencephalography (EEG). Also, we developed a real-time health 4.0 (H4.0) architecture for brain-controlled internet of things (IoT) enabled environments (BCE), which uses the classified MI task to assist disabled persons in controlling IoT-enabled environments such as lighting and heating, ventilation, and air-conditioning (HVAC). The first method for classification involves a simple and low-complex classification framework using a combination of regularized Riemannian mean (RRM) and linear SVM. Although this method performs better compared to state-of-the-art techniques, it still suffers from a nonnegligible misclassification rate. Hence, to overcome this, the second method offers a persistent decision engine (PDE) for the MI classification, which improves classification accuracy (CA) significantly. The proposed methods are validated using an in-house recorded four-class MI data set (data set I, collected over 14 subjects), and a four-class MI data set 2a of BCI competition IV (data set II, collected over 9 subjects). The proposed RRM architecture obtained average CAs of 74.30% and 67.60% when validated using datasets I and II, respectively. When analyzed along with the proposed PDE classification framework, an average CA of 92.25% on 12 subjects of data set I and 82.54% on 7 subjects of data set II is obtained. The results show that the PDE algorithm is more reliable for the classification of four-class MI and is also feasible for BCE applications. The proposed low-complex BCE architecture is implemented in real time using Raspberry Pi 3 Model B+ along with the Virgo EEG data acquisition system. The hardware

implementation results show that the proposed system architecture is well suited for body-wearable devices in the scenario of Health 4.0. We strongly feel that this study can aid in driving the future scope of BCE research.

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66. A Minimum Free Energy Model of Motor Learning

Brian Mitchell, Nina Lauharatanahirun, Jo Garcia, Nicholas F. Wymbs, Scott T. Grafton, Jean M. Vettel, Linda Petzold

https://doi.org/10.1162/neco_a_01219 | pp. 1945-1963

Even highly trained behaviors demonstrate variability, which is correlated with performance on current and future tasks. An objective of motor learning that is general enough to explain these phenomena has not been precisely formulated. In this six-week longitudinal learning study, participants practiced a set of motor sequences each day, and neuroimaging data were collected on days 1, 14, 28, and 42 to capture the neural correlates of the learning process. In our analysis, we first modeled the underlying neural and behavioral dynamics during learning. Our results demonstrate that the densities of whole-brain response, task-active regional response, and behavioral performance evolve according to a Fokker-Planck equation during the acquisition of a motor skill. We show that this implies that the brain concurrently optimizes the entropy of a joint density over neural response and behavior (as measured by sampling over multiple trials and subjects) and the expected performance under this density; we call this formulation of learning minimum free energy learning (MFEL). This model provides an explanation as to how behavioral variability can be tuned while simultaneously improving performance during learning. We then develop a novel variant of inverse reinforcement learning to retrieve the cost function optimized by the brain during the learning process, as well as the parameter used to tune variability. We show that this population-level analysis can be used to derive a learning objective that each subject optimizes during his or her study. In this way, MFEL effectively acts as a unifying principle, allowing users to precisely formulate learning objectives and infer their structure.

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67. A Mechanism for Synaptic Copy Between Neural Circuits

Yuxiu Shao, Binxu Wang, Andrew Sornborger, Louis Tao

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Cortical oscillations are central to information transfer in neural systems. Significant evidence supports the idea that coincident spike input can allow the neural threshold to be overcome and spikes to be propagated downstream in a circuit. Thus, an observation of oscillations in neural circuits would be an indication that repeated synchronous spiking may be enabling information transfer. However, for memory transfer, in which synaptic weights must be being transferred from one neural circuit (region) to another, what is the mechanism? Here, we present a synaptic transfer mechanism whose structure provides some understanding of the phenomena that have been implicated in memory transfer, including nested oscillations at various frequencies. The circuit is based on the principle of pulse-gated, graded information transfer between neural populations.

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68. One Step Back, Two Steps Forward: Interference and Learning in Recurrent Neural Networks

Chen Beer, Omri Barak

https://doi.org/10.1162/neco_a_01222 | pp. 1985-2003

Artificial neural networks, trained to perform cognitive tasks, have recently been used as models for neural recordings from animals performing these tasks. While some progress has been made in performing such comparisons, the evolution of network dynamics throughout learning remains unexplored. This is paralleled by an experimental focus on recording from trained animals, with few studies following neural activity throughout training. In this work, we address this gap in the realm of artificial networks by analyzing networks that are trained to perform memory and pattern generation tasks. The functional aspect of these tasks corresponds to dynamical objects in the fully trained network-a line attractor or a set of limit cycles for the two respective tasks. We use these dynamical objects as anchors to study the effect of learning on their emergence. We find that the sequential nature of learning-one trial at a time-has major consequences for the learning trajectory and its final outcome. Specifically, we show that least mean squares (LMS), a simple gradient descent suggested as a biologically plausible version of the FORCE algorithm, is constantly obstructed by forgetting, which is manifested as the destruction of dynamical objects from previous trials. The degree of interference is determined by the correlation between different trials. We show which specific ingredients of FORCE avoid this phenomenon. Overall, this difference results in convergence that is orders of magnitude slower for LMS. Learning implies accumulating information across multiple trials to form the overall concept of the task. Our results show that interference between trials can greatly affect learning in a learning-rule-dependent manner. These insights can help design experimental protocols that minimize such interference, and possibly infer underlying learning rules by observing behavior and neural activity throughout learning.

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69. Machine Learning of Time Series Using Time-Delay Embedding and Precision Annealing

Alexander J. A. Ty, Zheng Fang, Rivver A. Gonzalez, Paul J. Rozdeba, Henry D. I. Abarbanel

https://doi.org/10.1162/neco_a_01224 | pp. 2004-2024

Tasking machine learning to predict segments of a time series requires estimating the parameters of a ML model with input/output pairs from the time series. We borrow two techniques used in statistical data assimilation in order to accomplish this task: time-delay embedding to prepare our input data and precision annealing as a training method. The precision annealing approach identifies the global minimum of the action ([Formula: see text]). In this way, we are able to identify the number of training pairs required to produce good generalizations (predictions) for the time series. We proceed from a scalar time series [Formula: see text] and, using methods of nonlinear time series analysis, show how to produce a [Formula: see text]-dimensional time-delay embedding space in which the time series has no false neighbors as does the observed [Formula: see text] time series. In that [Formula: see text]-dimensional space, we explore the use of feedforward multilayer perceptrons as network models operating on [Formula: see text]-dimensional input and producing [Formula: see text]-dimensional outputs.

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70. A Novel Predictive-Coding-Inspired Variational RNN Model for Online Prediction and Recognition

Ahmadreza Ahmadi, Jun Tani

https://doi.org/10.1162/neco_a_01228 | pp. 2025-2074

This study introduces PV-RNN, a novel variational RNN inspired by predictive-coding ideas. The model learns to extract the probabilistic structures hidden in fluctuating temporal patterns by dynamically changing the stochasticity of its latent states. Its architecture attempts to address two major concerns of variational Bayes RNNs: how latent variables can learn meaningful representations and how the inference model can transfer future observations to the latent variables. PV-RNN does both by introducing adaptive vectors mirroring the training data, whose values can then be adapted differently during evaluation. Moreover, prediction errors during backpropagation-rather than external inputs during the forward computation-are used to convey information to the network about the external data. For testing, we introduce error regression for predicting unseen sequences as inspired by predictive coding that leverages those mechanisms. As in other variational Bayes RNNs, our model learns by maximizing a lower bound on the marginal likelihood of the sequential data, which is composed of two terms: the negative of the expectation of prediction errors and the negative of the Kullback-Leibler divergence between the prior and the approximate posterior distributions. The model introduces a weighting parameter, the meta-prior, to balance the optimization pressure placed on those two terms. We test the model on two data sets with probabilistic structures and show that with high values of the meta-prior, the network develops deterministic chaos through which the randomness of the data is imitated. For low values, the model behaves as a random process. The network performs best on intermediate values and is able to capture the latent probabilistic structure with good generalization. Analyzing the meta-prior's impact on the network allows us to precisely study the theoretical value and practical benefits of incorporating stochastic dynamics in our model. We demonstrate better prediction performance on a robot imitation task with our model using error regression compared to a standard variational Bayes model lacking such a procedure.

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71. On Functions Computed on Trees

Roozbeh Farhoodi, Khashayar Filom, Ilenna Simone Jones, Konrad Paul Kording

https://doi.org/10.1162/neco_a_01231 | pp. 2075-2137

Any function can be constructed using a hierarchy of simpler functions through compositions. Such a hierarchy can be characterized by a binary rooted tree. Each node of this tree is associated with a function that takes as inputs two numbers from its children and produces one output. Since thinking about functions in terms of computation graphs is becoming popular, we may want to know which functions can be implemented on a given tree. Here, we describe a set of necessary constraints in the form of a system of nonlinear partial differential equations that must be satisfied. Moreover, we prove that these conditions are sufficient in contexts of analytic and bit-valued functions. In the latter case, we explicitly enumerate discrete functions and observe that there are relatively few. Our point of view allows us to compare different neural network architectures in regard to their function spaces. Our work connects the structure of computation graphs with the functions they can implement and has potential applications to neuroscience and computer science.

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72. Integrating Flexible Normalization into Midlevel Representations of Deep Convolutional Neural Networks

Luis G. Sánchez Giraldo, Odelia Schwartz

https://doi.org/10.1162/neco_a_01226 | pp. 2138-2176

Deep convolutional neural networks (CNNs) are becoming increasingly popular models to predict neural responses in visual cortex. However, contextual effects, which are prevalent in neural processing and in perception, are not explicitly handled by current CNNs, including those used for neural prediction. In primary visual cortex, neural responses are modulated by stimuli spatially surrounding the classical receptive field in rich ways. These effects have been modeled with divisive normalization approaches, including flexible models, where spatial normalization is recruited only to the degree that responses from center and surround locations are deemed statistically dependent. We propose a flexible normalization model applied to midlevel representations of deep CNNs as a tractable way to study contextual normalization mechanisms in midlevel cortical areas. This approach captures nontrivial spatial dependencies among midlevel features in CNNs, such as those present in textures and other visual stimuli, that arise from tiling high-order features geometrically. We expect that the proposed approach can make predictions about when spatial normalization might be recruited in midlevel cortical areas. We also expect this approach to be useful as part of the CNN tool kit, therefore going beyond more restrictive fixed forms of normalization.

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73. Capturing the Forest but Missing the Trees: Microstates Inadequate for Characterizing Shorter-Scale EEG Dynamics

Saurabh Bhaskar Shaw, Kiret Dhindsa, J.P. Reilly, Suzanna Becker

https://doi.org/10.1162/neco_a_01229 | pp. 2177-2211

The brain is known to be active even when not performing any overt cognitive tasks, and often it engages in involuntary mind wandering. This resting state has been extensively characterized in terms of fMRI-derived brain networks. However, an alternate method has recently gained popularity: EEG microstate analysis. Proponents of microstates postulate that the brain discontinuously switches between four quasi-stable states defined by specific EEG scalp topologies at peaks in the global field potential (GFP). These microstates are thought to be "atoms of thought," involved with visual, auditory, salience, and attention processing. However, this method makes some major assumptions by excluding EEG data outside the GFP peaks and then clustering the EEG scalp topologies at the GFP peaks, assuming that only one microstate is active at any given time. This study explores the evidence surrounding these assumptions by studying the temporal dynamics of microstates and its clustering space using tools from dynamical systems analysis, fractal, and chaos theory to highlight the shortcomings in microstate analysis. The results show evidence of complex and chaotic EEG dynamics outside the GFP peaks, which is being missed by microstate analysis. Furthermore, the winner-takes-all approach of only one microstate being active at a time is found to be inadequate since the dynamic EEG scalp topology does not always resemble that of the assigned microstate, and there is competition among the different microstate classes. Finally, clustering space analysis shows that the four microstates do not cluster into four distinct and separable clusters. Taken collectively, these results show that the discontinuous description of EEG microstates is inadequate when looking at nonstationary short-scale EEG dynamics.

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74. Dynamic Integrative Synaptic Plasticity Explains the Spacing Effect in the Transition from Short- to Long-Term Memory

Terry Elliott

https://doi.org/10.1162/neco_a_01227 | pp. 2212-2251

Repeated stimuli that are spaced apart in time promote the transition from short- to long-term memory, while massing repetitions together does not. Previously, we showed that a model of integrative synaptic plasticity, in which plasticity induction signals are integrated by a low-pass filter before plasticity is expressed, gives rise to a natural timescale at which to repeat stimuli, hinting at a partial account of this spacing effect. The account was only partial because the important role of neuromodulation was not considered. We now show that by extending the model to allow dynamic integrative synaptic plasticity, the model permits synapses to robustly discriminate between spaced and massed repetition protocols, suppressing the response to massed stimuli while maintaining that to spaced stimuli. This is achieved by dynamically coupling the filter decay rate to neuromodulatory signaling in a very simple model of the signaling cascades downstream from cAMP production. In particular, the model's parameters may be interpreted as corresponding to the duration and amplitude of the waves of activity in the MAPK pathway. We identify choices of parameters and repetition times for stimuli in this model that optimize the ability of synapses to discriminate between spaced and massed repetition protocols. The model is very robust to reasonable changes around these optimal parameters and times, but for large changes in parameters, the model predicts that massed and spaced stimuli cannot be distinguished or that the responses to both patterns are suppressed. A model of dynamic integrative synaptic plasticity therefore explains the spacing effect under normal conditions and also predicts its breakdown under abnormal conditions.

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75. Mutual Inhibition with Few Inhibitory Cells via Nonlinear Inhibitory Synaptic Interaction

Felix Weissenberger, Marcelo Matheus Gauy, Xun Zou, Angelika Steger

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In computational neural network models, neurons are usually allowed to excite some and inhibit other neurons, depending on the weight of their synaptic connections. The traditional way to transform such networks into networks that obey Dale's law (i.e., a neuron can either excite or inhibit) is to accompany each excitatory neuron with an inhibitory one through which inhibitory signals are mediated. However, this requires an equal number of excitatory and inhibitory neurons, whereas a realistic number of inhibitory neurons is much smaller. In this letter, we propose a model of nonlinear interaction of inhibitory synapses on dendritic compartments of excitatory neurons that allows the excitatory neurons to mediate inhibitory signals through a subset of the inhibitory population. With this construction, the number of required inhibitory neurons can be reduced tremendously.

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76. Adversarial Feature Alignment: Avoid Catastrophic Forgetting in Incremental Task Lifelong Learning

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, is one of the major roadblocks that prevent deep neural networks from achieving human-level artificial intelligence. Several research efforts (e.g., lifelong or continual learning algorithms) have proposed to tackle this problem. However, they either suffer from an accumulating drop in performance as the task sequence

grows longer, or require storing an excessive number of model parameters for historical memory, or cannot obtain competitive performance on the new tasks. In this letter, we focus on the incremental multitask image classification scenario. Inspired by the learning process of students, who usually decompose complex tasks into easier goals, we propose an adversarial feature alignment method to avoid catastrophic forgetting. In our design, both the low-level visual features and high-level semantic features serve as soft targets and guide the training process in multiple stages, which provide sufficient supervised information of the old tasks and help to reduce forgetting. Due to the knowledge distillation and regularization phenomena, the proposed method gains even better performance than fine-tuning on the new tasks, which makes it stand out from other methods. Extensive experiments in several typical lifelong learning scenarios demonstrate that our method outperforms the state-of-the-art methods in both accuracy on new tasks and performance preservation on old tasks.

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77. Every Local Minimum Value Is the Global Minimum Value of Induced Model in Nonconvex Machine Learning

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For nonconvex optimization in machine learning, this article proves that every local minimum achieves the globally optimal value of the perturbable gradient basis model at any differentiable point. As a result, nonconvex machine learning is theoretically as supported as convex machine learning with a handcrafted basis in terms of the loss at differentiable local minima, except in the case when a preference is given to the handcrafted basis over the perturbable gradient basis. The proofs of these results are derived under mild assumptions. Accordingly, the proven results are directly applicable to many machine learning models, including practical deep neural networks, without any modification of practical methods. Furthermore, as special cases of our general results, this article improves or complements several state-of-the-art theoretical results on deep neural networks, deep residual networks, and overparameterized deep neural networks with a unified proof technique and novel geometric insights. A special case of our results also contributes to the theoretical foundation of representation learning.

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78. Can Grid Cell Ensembles Represent Multiple Spaces?

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The way grid cells represent space in the rodent brain has been a striking discovery, with theoretical implications still unclear. Unlike hippocampal place cells, which are known to encode multiple, environment-dependent spatial maps, grid cells have been widely believed to encode space through a single low-dimensional manifold, in which coactivity relations between different neurons are preserved when the environment is changed. Does it have to be so? Here, we compute, using two alternative mathematical models, the storage capacity of a population of grid-like units, embedded in a continuous attractor neural network, for multiple spatial maps. We show that distinct representations of multiple environments can coexist, as existing models for grid cells have the potential to express several sets of hexagonal grid patterns, challenging the view of a universal grid map. This suggests that a population of grid cells can encode multiple noncongruent metric relationships, a feature that could in principle allow a grid-like code to

represent environments with a variety of different geometries and possibly conceptual and cognitive spaces, which may be expected to entail such context-dependent metric relationships.

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79. Replicating Neuroscience Observations on ML/MF and AM Face Patches by Deep Generative Model

Tian Han, Xianglei Xing, Jiawen Wu, Ying Wu

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paper (Chang & Tsao, 2017) reports an interesting discovery. For the face stimuli generated by a pretrained active appearance model (AAM), the responses of neurons in the areas of the primate brain that are responsible for face recognition exhibit a strong linear relationship with the shape variables and appearance variables of the AAM that generates the face stimuli. In this letter, we show that this behavior can be replicated by a deep generative model, the generator network, that assumes that the observed signals are generated by latent random variables via a top-down convolutional neural network. Specifically, we learn the generator network from the face images generated by a pretrained AAM model using a variational autoencoder, and we show that the inferred latent variables of the learned generator network have a strong linear relationship with the shape and appearance variables of the AAM model that generates the face images. Unlike the AAM model, which has an explicit shape model where the shape variables generate the control points or landmarks, the generator network has no such shape model and shape variables. Yet it can learn the shape knowledge in the sense that some of the latent variables of the learned generator network capture the shape variations in the face images generated by AAM.

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80. Reinforcement Learning in Spiking Neural Networks with Stochastic and Deterministic Synapses

Mengwen Yuan, Wu Xi, Rui Yan, Huajin Tang

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Though succeeding in solving various learning tasks, most existing reinforcement learning (RL) models have failed to take into account the complexity of synaptic plasticity in the neural system. Models implementing reinforcement learning with spiking neurons involve only a single plasticity mechanism. Here, we propose a neural realistic reinforcement learning model that coordinates the plasticities of two types of synapses: stochastic and deterministic. The plasticity of the stochastic synapse is achieved by the hedonistic rule through modulating the release probability of synaptic neurotransmitter, while the plasticity of the deterministic synapse is achieved by a variant of a reward-modulated spike-timing-dependent plasticity rule through modulating the synaptic strengths. We evaluate the proposed learning model on two benchmark tasks: learning a logic gate function and the 19-state random walk problem. Experimental results show that the coordination of diverse synaptic plasticities can make the RL model learn in a rapid and stable form.

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81. Bayesian Filtering with Multiple Internal Models: Toward a Theory of Social Intelligence

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To exhibit social intelligence, animals have to recognize whom they are communicating with. One way to make this inference is to select among internal generative models of each conspecific who may be encountered. However, these models also have to be learned via some form of Bayesian belief updating. This induces an interesting problem: When receiving sensory input generated by a particular conspecific, how does an animal know which internal model to update? We consider a theoretical and neurobiologically plausible solution that enables inference and learning of the processes that generate sensory inputs (e.g., listening and understanding) and reproduction of those inputs (e.g., talking or singing), under multiple generative models. This is based on recent advances in theoretical neurobiology-namely, active inference and post hoc (online) Bayesian model selection. In brief, this scheme fits sensory inputs under each generative model. Model parameters are then updated in proportion to the probability that each model could have generated the input (i.e., model evidence). The proposed scheme is demonstrated using a series of (real zebra finch) birdsongs, where each song is generated by several different birds. The scheme is implemented using physiologically plausible models of birdsong production. We show that generalized Bayesian filtering, combined with model selection, leads to successful learning across generative models, each possessing different parameters. These results highlight the utility of having multiple internal models when making inferences in social environments with multiple sources of sensory information.

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82. Safe Triplet Screening for Distance Metric Learning

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Distance metric learning has been widely used to obtain the optimal distance function based on the given training data. We focus on a triplet-based loss function, which imposes a penalty such that a pair of instances in the same class is closer than a pair in different classes. However, the number of possible triplets can be quite large even for a small data set, and this considerably increases the computational cost for metric optimization. In this letter, we propose safe triplet screening that identifies triplets that can be safely removed from the optimization problem without losing the optimality. In comparison with existing safe screening studies, triplet screening is particularly significant because of the huge number of possible triplets and the semidefinite constraint in the optimization problem. We demonstrate and verify the effectiveness of our screening rules by using several benchmark data sets.

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83. The Effect of Signaling Latencies and Node Refractory States on the Dynamics of Networks

Gabriel A. Silva

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We describe the construction and theoretical analysis of a framework derived from canonical neurophysiological principles that model the competing dynamics of incident signals into nodes along directed edges in a network. The framework describes the dynamics between the offset in the latencies of propagating signals, which reflect the geometry of the edges and conduction velocities, and the internal refractory dynamics and processing times of the downstream node receiving the signals. This framework

naturally extends to the construction of a perceptron model that takes into account such dynamic geometric considerations. We first describe the model in detail, culminating with the model of a geometric dynamic perceptron. We then derive upper and lower bounds for a notion of optimal efficient signaling between vertex pairs based on the structure of the framework. Efficient signaling in the context of the framework we develop here means that there needs to be a temporal match between the arrival time of the signals relative to how quickly nodes can internally process signals. These bounds reflect numerical constraints on the compensation of the timing of signaling events of upstream nodes attempting to activate downstream nodes they connect into that preserve this notion of efficiency. When a mismatch between signal arrival times and the internal states of activated nodes occurs, it can cause a breakdown in the signaling dynamics of the network. In contrast to essentially all of the current state of the art in machine learning, this work provides a theoretical foundation for machine learning and intelligence architectures based on the timing of node activations and their abilities to respond rather than necessary changes in synaptic weights. At the same time, the theoretical ideas we developed are guiding the discovery of experimentally testable new structure-function principles in the biological brain.

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84. Spike-Based Winner-Take-All Computation: Fundamental Limits and Order-Optimal Circuits

Lili Su, Chia-Jung Chang, Nancy Lynch

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Winner-take-all (WTA) refers to the neural operation that selects a (typically small) group of neurons from a large neuron pool. It is conjectured to underlie many of the brain's fundamental computational abilities. However, not much is known about the robustness of a spike-based WTA network to the inherent randomness of the input spike trains. In this work, we consider a spike-based [Formula: see text]–WTA model wherein [Formula: see text] randomly generated input spike trains compete with each other based on their underlying firing rates and [Formula: see text] winners are supposed to be selected. We slot the time evenly with each time slot of length 1 ms and model the [Formula: see text] input spike trains as [Formula: see text] independent Bernoulli processes. We analytically characterize the minimum waiting time needed so that a target minimax decision accuracy (success probability) can be reached. We first derive an information-theoretic lower bound on the waiting time. We show that to guarantee a (minimax) decision error [Formula: see text] (where [Formula: see text]), the waiting time of any WTA circuit is at least [Formula: see text]where [Formula: see text] is a finite set of rates and [Formula: see text] is a difficulty parameter of a WTA task with respect to set [Formula: see text] for independent input spike trains. Additionally, [Formula: see text] is independent of [Formula: see text], [Formula: see text], and [Formula: see text]. We then design a simple WTA circuit whose waiting time is [Formula: see text]provided that the local memory of each output neuron is sufficiently long. It turns out that for any fixed [Formula: see text], this decision time is order-optimal (i.e., it matches the above lower bound up to a multiplicative constant factor) in terms of its scaling in [Formula: see text], [Formula: see text], and [Formula: see text].

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85. On the Effect of the Activation Function on the Distribution of Hidden Nodes in a Deep Network

Philip M. Long, Hanie Sedghi

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We analyze the joint probability distribution on the lengths of the vectors of hidden variables in different layers of a fully connected deep network, when the weights and biases are chosen randomly according to gaussian distributions. We show that if the activation function [Formula: see text] satisfies a minimal set of assumptions, satisfied by all activation functions that we know that are used in practice, then, as the width of the network gets large, the “length process” converges in probability to a length map that is determined as a simple function of the variances of the random weights and biases and the activation function [Formula: see text]. We also show that this convergence may fail for [Formula: see text] that violate our assumptions. We show how to use this analysis to choose the variance of weight initialization, depending on the activation function, so that hidden variables maintain a consistent scale throughout the network.